

## 11 Passive Shimming for MR450

### 11.1 Personnel Requirements

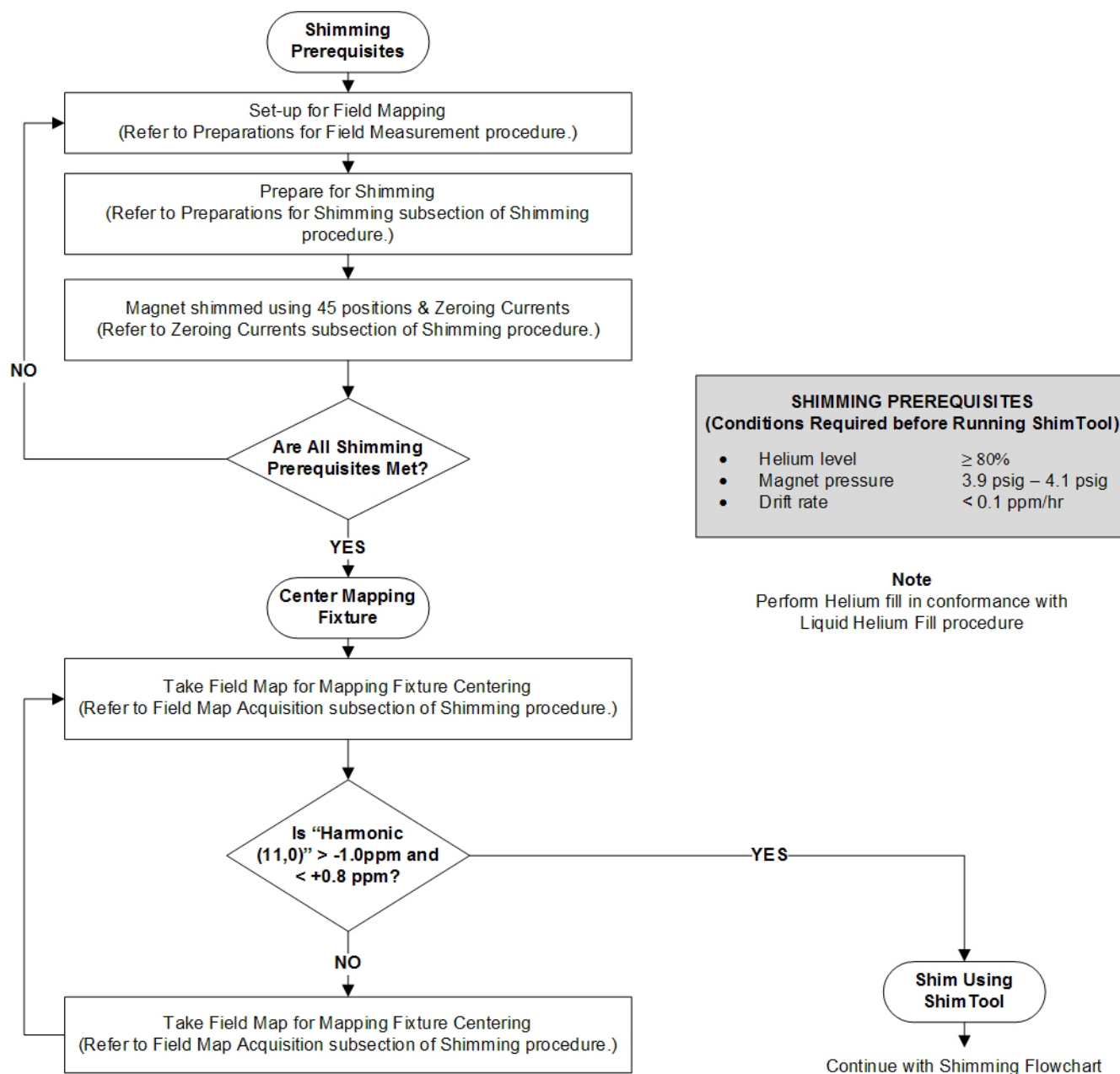
| Personnel Requirements | Preliminary Reqs      | Procedure                                  | Finalization          |
|------------------------|-----------------------|--------------------------------------------|-----------------------|
| 2                      | Included in Procedure | 10 hours (excluding Field Stability Check) | Included in Procedure |

### 11.2 Overview

Shimming is the process of measuring the magnet's field over a fixed volume, then calculating and installing active (superconducting) and passive (steel) shims to improve field homogeneity. Perform on-site Shimming to compensate for unequal distribution of room steel that is affecting field homogeneity. Additional Shimming may be required following any significant environmental changes near the magnet. It is important to perform Shimming after the magnet is sited and ramped with the magnetic field stabilized.

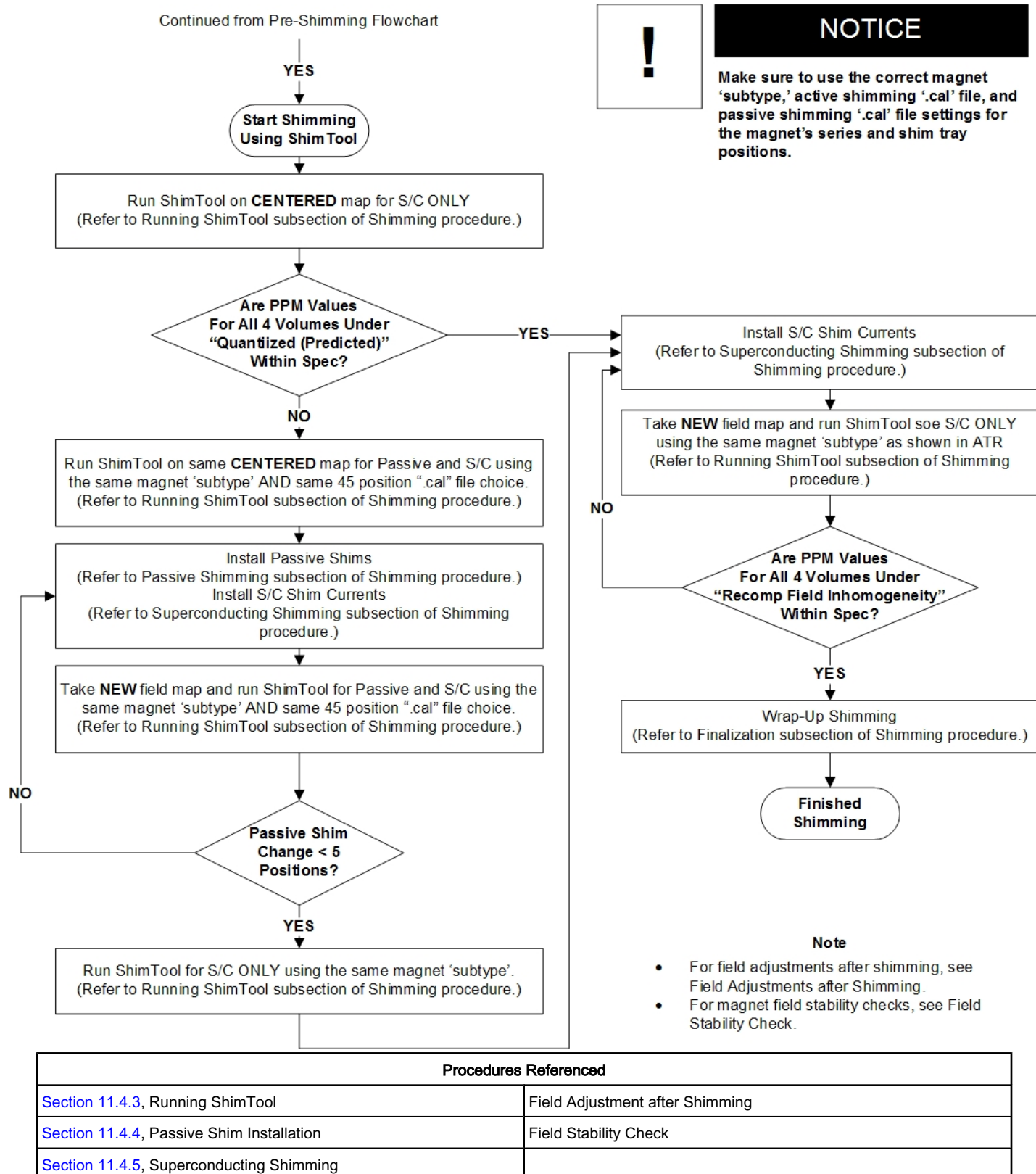
- NOTE:** After reapplying factory Superconducting (S/C) Shim Currents to the magnet:
- Use the ShimTool software to calculate any additional passive and S/C (active) shimming required on site. Flowcharts of shimming prerequisites and the shimming process are shown in and [Illustration 3-148](#), respectively.
  - Allow the magnet to stabilize to <0.1 ppm/hr. main field drift before shimming (takes 12 hours). Perform other commissioning functions during this time.
  - Make sure the minimum helium level to start shimming is 85%.
  - Always use the most **recent** revision of GE Shimtool Software CD 5102783. The LCC Shimming procedure stated here includes shimming of the RB series LCC magnets.

Illustration 3-147: Pre-Shimming Flowchart



| Procedures Referenced                              |                                                           |
|----------------------------------------------------|-----------------------------------------------------------|
| Liquid Helium Fill                                 | <a href="#">Section 11.4.1, Preparations for Shimming</a> |
| <a href="#">Preparations for Field Measurement</a> | <a href="#">Section 11.4.2, Zeroing Currents</a>          |
|                                                    | <a href="#">Section 11.4.3.1, Field Map Acquisition</a>   |

Illustration 3-148: Shimming Flowchart



| Procedures Referenced |  |
|-----------------------|--|
| Finalization, Step 1  |  |

## 11.3 Preliminary Requirements

### 11.3.1 Tools and Test Equipment

| Item                                                                                                                           | Qty                    | Effectivity | Part#                 | Manufacturer |
|--------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------|-----------------------|--------------|
| 1.5T Warning Sign & Label Kit                                                                                                  | 1                      | -           | 46-258770G4           | -            |
| Portable Oxygen Monitor                                                                                                        | 1/<br>perso<br>n       | -           | 2287000               | -            |
| Cryogen Safety Wear Kit                                                                                                        | 1 kit/<br>perso<br>n   | -           | 46-271137G1           | -            |
| Nonabsorbent Protective Clothing (long sleeve shirt and pants)                                                                 | 1 set/<br>perso<br>n   | -           | -                     | -            |
| Nonferrous Safety Shoes                                                                                                        | 1 pr./<br>perso<br>n   | -           | -                     | -            |
| Brass Master Padlock with Brass Shackle                                                                                        | 1                      | -           | 46-194427P320         | -            |
| Brass Master Transition Padlock                                                                                                | 1                      | -           | 2387081               | -            |
| Black & White on Red Lockout Tag                                                                                               | 1 packa<br>ge of<br>25 | -           | 46-194427P322         | -            |
| Transition LOTO Tag                                                                                                            | 1                      | -           | 46-194427P313         | -            |
| Line Cord Plug Cover, Plugs $\leq 3$ in. (76 mm) wide x $\leq 5.875$ in. (149 mm) long, Cords $\leq 1.25$ in. (31 mm) diameter | 1                      | -           | 46-194427P321         | -            |
| Shim Power Supply                                                                                                              | 1                      | -           | 46-260777G3           | -            |
| 750 Amp Ramp Cable Kit 2135435 or 1,000 Amp Ramp Cable Kit with Ground 2353394                                                 | 1                      | -           | 2135435 or<br>2353394 | -            |

| Item                                                                                                                                                     | Qty   | Effectivity                                         | Part#              | Manufacturer |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------|--------------------|--------------|
| Shim Cable Kit (if not already ordered with the magnet)                                                                                                  | 1     | -                                                   | 2135558            | -            |
| MetroLab Teslameter Kit                                                                                                                                  | 1     | -                                                   | 46-251865G4        | -            |
| No. 5 MetroLab Probe (included in MetroLab Teslameter Kit 46-251865G4)                                                                                   | 1     | -                                                   | 46-255839P104      | -            |
| LCC Shim Camera Kit                                                                                                                                      | 1     | -                                                   | 2386028-2          | -            |
| GE ShimTool Software CD 5102783 (included in LCC Shim Camera Kit 2386028-2)                                                                              | 1     | -                                                   | 5102783            | -            |
| HDv Passive Shimming Kit (includes Universal Passive Shimming Kit 2380758-2 and HDv Passive Shim Extraction Tool 5172460)                                | 1     | -                                                   | 5266042            | -            |
| HDv Field Passive Shim Kit                                                                                                                               | 1     | -                                                   | 5263916            | -            |
| Titanium Tool Kit, Basic Set 5113258; Titanium Tool Kit, Installation Set 5112581; or other non-ferromagnetic tools (magnets with magnetic fields >1.5T) | 1 Kit | -                                                   | 5113258 or 5112581 | -            |
| HD Service Platform                                                                                                                                      | 1     | HD, HD <sub>e</sub> , HD <sub>x</sub> , MR355/MR360 | 5155291            | -            |
| MR355/MR360 Magnet Side Kit Protection Plate                                                                                                             | 1     | MR355/MR360                                         | 5366607            | -            |
| HDv Service Platform with Side Shield                                                                                                                    | 1     | MR450                                               | 5155291-2          | -            |
| Universal Service Ladder/Platform                                                                                                                        | 1     | Systems prior to HD                                 | 2319156            | -            |
| Service Ladder/Platform                                                                                                                                  | 1     | Fixed Site Systems prior to HD                      | 2230707            | -            |
| Service Ladder/Platform (Service space <30 in. or 762 mm)                                                                                                | 1     | Mobile Site Systems prior to HD                     | 2230707-2          | -            |

| Item                                                                                                                                                                                                                      | Qty | Effectivity | Part# | Manufacturer |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------|-------|--------------|
| Service Methods documentation for the magnet/enclosures (available through the Common Documentation Library at <a href="http://gehealthcare.com">gehealthcare.com</a> or your local GE Healthcare Service Representative) | 1   | -           | -     | -            |

### 11.3.2 Consumables

| Item                                                                                                                                                                                            | Qty | Effectivity | Part# | Manufacturer |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------|-------|--------------|
| Optional Field-Supplied Item: 10 mil thick flexible plastic sheets (X-ray film, Mylar or equivalent), 10 in. x 14 in. (250 mm x 350 mm), minimum. Must be free of sharp edges and bent corners. | 1   | -           | -     | -            |

### 11.3.3 Safety



#### **⚠ DANGER**

POTENTIAL FATAL INJURY!

BEFORE AND WHILE SHIMMING THE MAGNET:

- MAKE SURE TO REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS, AND
- MAKE SURE TO FULLY COMPLY WITH ALL REQUIRED ITEMS FOR SHIMMING IN THE MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS SUBSECTION OF [CHAPTER 1, MR MAGNET SAFETY REQUIREMENTS](#).



#### **⚠ DANGER**

POTENTIAL FATAL INJURY!

BEFORE SHIMMING THE MAGNET:

HAVE ALL WORK ASSISTANTS OR WORK OBSERVERS COMPLY WITH THE BUDDY SYSTEM REQUIREMENTS & CERTIFICATION SUBSECTION OF "MR MAGNET - SAFETY REQUIREMENTS."

**⚠ WARNING**

PERSONAL INJURY AND EQUIPMENT DAMAGE  
STRONG MAGNETIC FIELD

WHEN SERVICING ANY MAGNETIC EQUIPMENT, IT IS CRITICALLY IMPORTANT THAT THE SERVICE ENGINEER CONSCIOUSLY PLAN THE PATH TO BE TAKEN WHEN MOVING HIGHLY FERROUS DEVICES IN THE MAGNET ENVIRONMENT. THE PATH SHOULD BE AS FAR FROM THE MAGNET AS PRACTICAL AND AVOID HIGH FLUX-DENSITY FIELDS.

SAFETY REQUIREMENTS

- MOVEMENT OF FERROUS MATERIAL IN THE MAGNET ROOM MUST FOLLOW THE GE SERVICE PROCEDURE FOR THAT DEVICE. WHEN EXITING, MOVE AWAY FROM THE MAGNET IN THE MOST DIRECT MANNER POSSIBLE. PASSIVE SHIM TRAY MUST BE INSERTED/ REMOVED WITH THE SHIM TRAY POINTED TOWARDS THE MAGNET. THE SERVICE PERSONNEL SHOULD NEVER BE BETWEEN THE SHIM TRAY AND THE MAGNET. NOR SHOULD ANY SERVICE PERSONNEL BE ON THE SERVICE END OF THE MAGNET WHEN TRAYS ARE BEING INSERTED.
- TWO (2) MR SAFETY TRAINED PERSONNEL MUST BE PRESENT AT ALL TIMES WHEN SERVICING HIGHLY FERROUS DEVICES IN THE AREAS OF MAGNETIC FIELDS.

WHEN PLANNING A SERVICE PATH, IT IS CRITICAL THAT THE PATH BE CLEAR AND SUFFICIENTLY WIDE. ENSURE THAT THERE ARE NO TRIP HAZARDS, OBSTACLES, CLUTTER, SLIPPERY SURFACES OR OTHER ITEMS EVEN PARTIALLY RESTRICTING THE PATH. IF THERE ARE PORTABLE OBSTACLES IN A PATH, REMOVE THEM FROM THE AREA AND REPLACE THEM AFTER THE SERVICE ACTION IS COMPLETED. IT IS REQUIRED TO WALK THE PATH PRIOR TO BEGINNING SERVICE TO ENSURE THAT THERE IS SUFFICIENT SPACE THROUGH WHICH TO PASS FOR YOURSELF AND THE OBJECT BEING SERVICED.

**⚠ WARNING**

ASPHYXIATION HAZARD!

THE SHIMMING PROCEDURE GENERATES ODORLESS, COLORLESS HELIUM GAS THAT DISPLACES OXYGEN IN THE AIR.

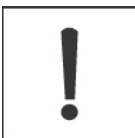
MAKE SURE MAGNET ROOM VENT EXHAUST FAN IS ON BEFORE SHIMMING THE MAGNET.

**⚠ WARNING**

ELECTRICAL SHOCK HAZARD!

CONTACT WITH CONNECTORS LEADING TO AN ENERGIZED POWER SUPPLY CAN CAUSE ELECTRICAL SHOCK.

FOLLOW [CHAPTER 1, OSHA LOCKOUT/TAGOUT](#) (LOTO) PROCEDURES WHILE MAKING ELECTRICAL CONNECTIONS TO A MAIN POWER SUPPLY OR A SHIM POWER SUPPLY.

**NOTICE**

Verify that Magnet Monitor is on and operational, and comply with the following:

- If using proprietary Magnet Monitor software, verify that the Magnet Monitor Pressure Control is enabled and operational.
- Moving articles or equipment in the Magnet Room may affect field readings. Do NOT move articles or equipment in the Magnet Room while adjusting shim currents.
- ONLY use nonmagnetic tools when working in the Magnet Room.

**NOTICE**

Service Platform with side shield must be installed when working on the magnet to protect side-mounted components from damage.

Illustration 3-149: Service Platform with Side Shield

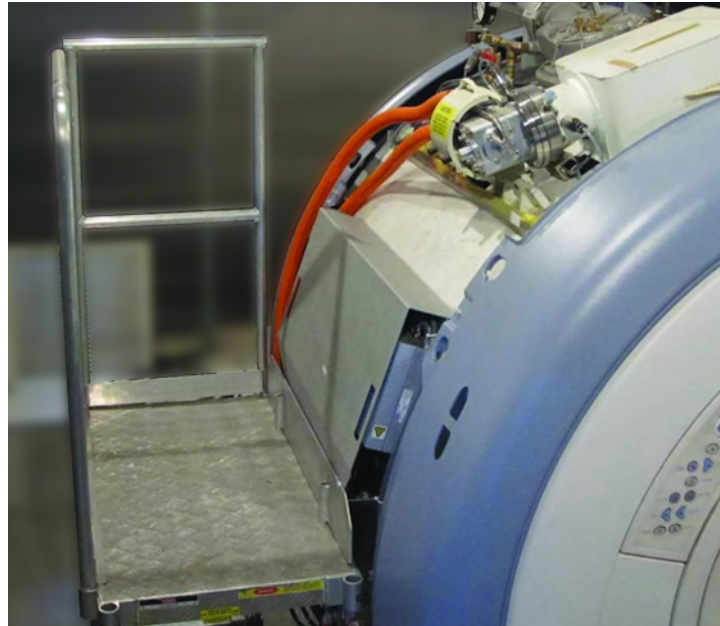




## NOTICE

For the MR355/MR360, the Service Platform (5155291) and Magnet Side Kit Protection Plate (5366607) must be used together when working on the magnet, to protect side-mounted components from damage. (See illustration below.)

Illustration 3-150: MR355/MR360 Service Platform & Magnet Side Kit Protection Plate



### 11.3.4 Required Conditions

| Condition                                                                                                                                                                                                                                                                                                                                                                                   | Reference                                                                | Effectivity |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-------------|
| Make sure the area is secure and that ALL required Warning Signs are posted to meet the safety requirements stated in the following document before starting Passive Shimming.                                                                                                                                                                                                              | <a href="#">Chapter 1, MR Magnet Safety Requirements</a>                 | -           |
| Removal of some magnet enclosure components may be required to complete this procedure. Refer to the following document and the appropriate Service Methods documentation (available through the Common Documentation Library at <a href="http://gehealthcare.com">gehealthcare.com</a> or your local GE Healthcare Service Representative) before continuing if cover removal is required. | <a href="#">Chapter 5, Introduction to HDe, HDx and MR450 Enclosures</a> | -           |
| Verify that the magnet helium level is $\geq 85\%$ . Fill the magnet in conformance with the following document to make sure the helium level is $\geq 85\%$ before shimming.                                                                                                                                                                                                               | Liquid Helium Fill                                                       | -           |

| Condition                                                                                                                                                                                                                                                                                                                                                                 | Reference | Effectivity |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|
| Verify that the magnet's magnetic field drift rate is <0.1 ppm/hr. in conformance with <a href="#">Section 11.4.1.3</a> , Field Stability Check before starting <a href="#">Section 11.4.3</a> , Running ShimTool.                                                                                                                                                        | -         | -           |
| Verify that Magnet Monitor is on and operational. If using proprietary Magnet Monitor software, verify that Magnet Monitor Pressure Control is enabled and operational. Verify that magnet pressure is steady at 3.9 to 4.1 psig in conformance with <a href="#">Section 11.4.1.4</a> , Magnet Monitor before starting <a href="#">Section 11.4.3</a> , Running ShimTool. | -         | -           |
| Make sure to re-establish the S/C Shim Currents in conformance with <a href="#">Section 11.4.2</a> , Zeroing Currents before starting <a href="#">Section 11.4.3</a> , Running ShimTool.                                                                                                                                                                                  | -         | -           |
| Make sure the Magnet Room exhaust fan is on and operational and the Magnet Room door is propped open before starting this procedure.                                                                                                                                                                                                                                      | -         | -           |

## 11.4 Procedure

### 11.4.1 Preparations for Shimming

#### 11.4.1.1 Initial Preparations

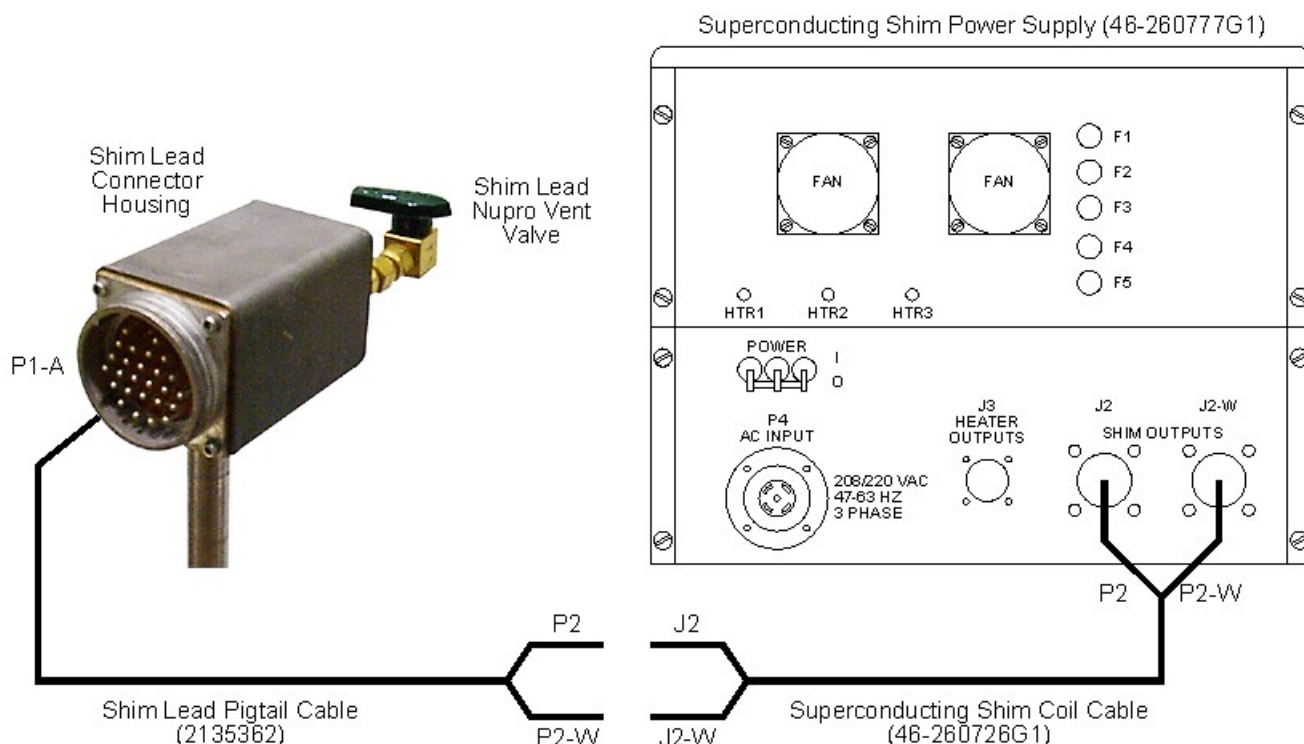
This section **must** be completed before shimming the magnet.

1. Install the Field Mapping Fixture if not already done. Refer to the *Universal Mapping Fixture 5266042 Set Up* section of [Preparations for Field Measurement](#) for details.
2. Allow the magnet to stabilize to <0.1 ppm/hr. main field drift before shimming (takes 12 hours). Perform other commissioning functions during this time.
3. Verify that the magnet helium level is ≥85%. Fill the magnet in conformance with Liquid Helium Fill to make sure the helium level is ≥85% before continuing.

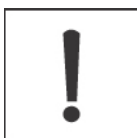
#### 11.4.1.2 Shim Coil Power Supply Connections

1. Verify that the Shim Lead Pigtail Cable 2135362 is connected to **P1-A** on the Shim Lead Assembly in conformance with the *S/C Shim Coil Power Supply Connections to Magnet* subsection of Connections for Ramping and Shimming and the illustration below.

Illustration 3-151: Shim Power Supply Connections to P1-A



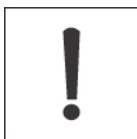
2. Make sure the Shim Lead Assembly is engaged in conformance with Shim Lead Engagement and Disengagement.
3. Open the Nupro Shim Lead Vent Valve to increase the Shim Leads' ability to carry current.



### NOTICE

Do NOT connect input power to the S/C Shim Coil Power Supply until it is verified that ALL current controls are set to zero (turned fully counterclockwise) and the supply is turned off.

Connecting the S/C Shim Coil Harness to the magnet when the S/C Shim Coil Power Supply is turned on can cause irreparable damage to the vapor-cooled shim leads.



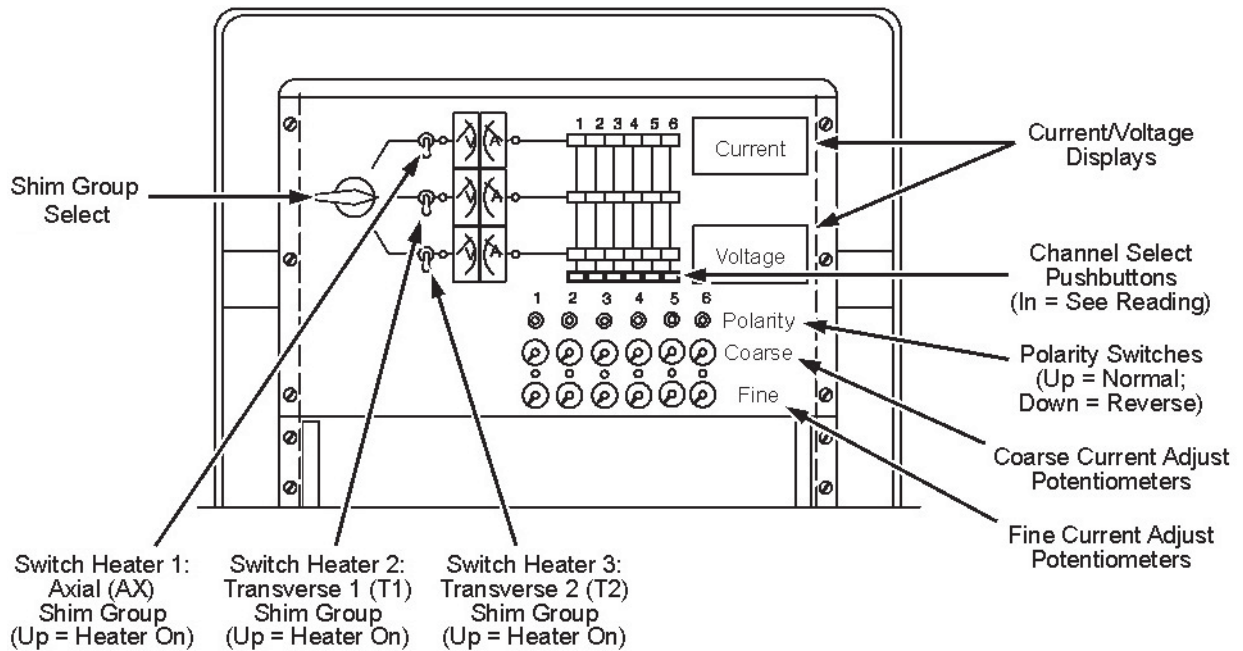
### NOTICE

Equipment damage is possible if heater currents are not set correctly.

4. Remove LOTO and connect input power to the S/C Shim Coil Power Supply. Make sure the Heater Currents are T1 = 610 mA ( $\pm 5$  mA), T2 = 610 mA ( $\pm 5$  mA), and Axial = 710 mA ( $\pm 10$  mA). See Section 4 of the vendor's service manual for details.

5. After sufficient cooling of the Shim Lead Assembly, verify that all six Shim Power Supply Channels operate at both positive and negative 250 mA for the Axial, Transverse 1, and Transverse 2 shim groups.

**Illustration 3-152: Shim Power Supply Controls (Front Panel)**

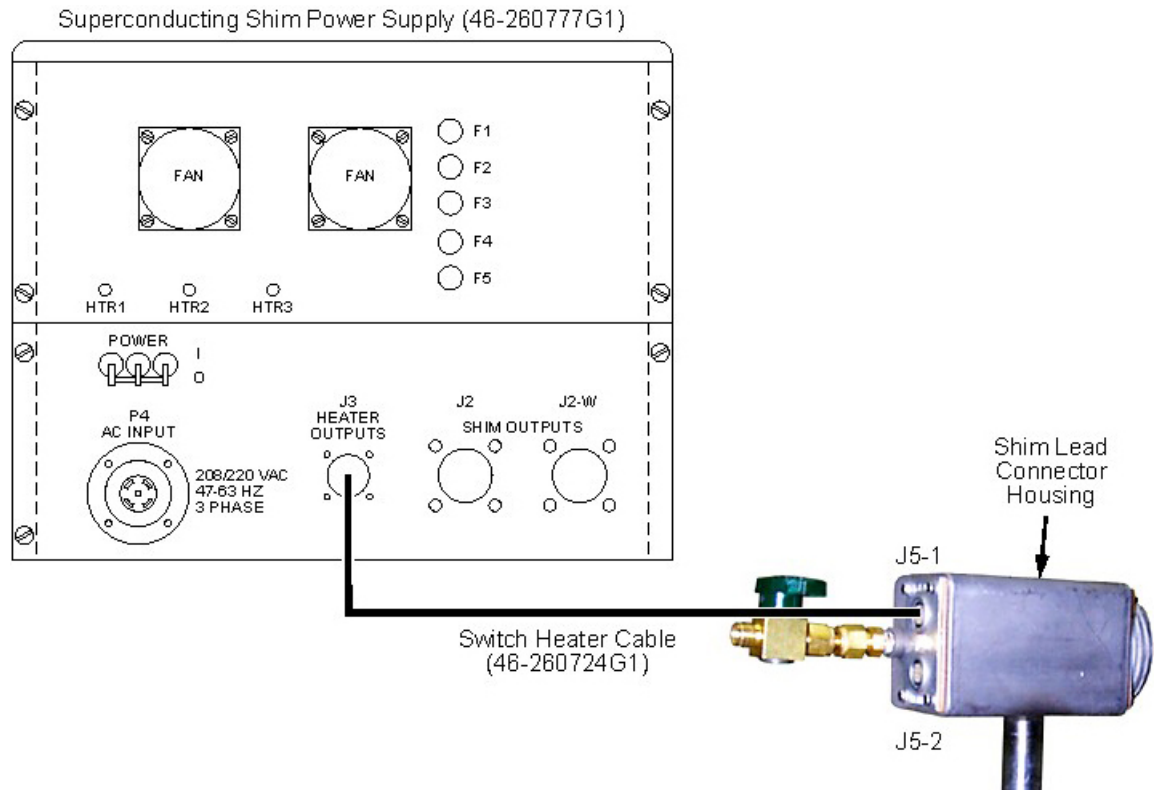


## WARNING

**ELECTRICAL SHOCK HAZARD!**  
CONTACT WITH CONNECTORS LEADING TO AN ENERGIZED POWER SUPPLY CAN CAUSE ELECTRICAL SHOCK.  
FOLLOW LOTO PROCEDURES WHILE MAKING ELECTRICAL CONNECTIONS TO THE SHIM POWER SUPPLY.

6. Disconnect and LOTO input power to the Shim Coil Power Supply. Verify that no voltage is present by using a DVM or equivalent measuring device.
7. Refer to [Illustration 3-151](#). Check that:
  - a. Connectors P2 and P2W on the S/C Shim Coil Cable 46-260726G1 (Run 603) are connected to the **J2** and **J2-W** Connectors (Shim Outputs) on the back of the Power Supply Cabinet (MS7-A1).
  - b. Connectors **J2** and **J2-W** on the other end of S/C Shim Coil Cable 46-260726G1 are connected to the Shim Lead Pigtail Cable 2135362.
8. Connect P703 on Switch Heater Cable 46-260724G1 (Run 604) to Connector **J3** (Heater Outputs) on the back of the S/C Shim Coil Power Supply Cabinet MS7-A1 as shown below.

Illustration 3-153: Shim Power Supply Connections to J5-1



## NOTICE

Connect only one Heater Wire Harness to the Shim Lead Assembly.

9. Check that Connector **P5** on the Switch Heater Cable (Run 604) is connected to Connector **J5-1** or **J5-2** on the Shim Lead Assembly.

### 11.4.1.3 Field Stability Check

The Field Stability Check requires 12 hours between the first and second measurements.

1. Verify that magnet pressure is 4.0 psig,  $\pm 0.1$  psig.
  - a. **Initial on-site Shimming:** Turn on the T1, T2, and Axial heaters as required to increase the magnet pressure to 4.0 psig,  $\pm 0.1$  psig.
  - b. **If any currents loaded (Reshimming):** Adjust the magnet pressure to 4.0 psig,  $\pm 0.1$  psig using the Magnet Monitor.



## NOTICE

Do NOT turn on the S/C Shim Coil Power Supply until it is verified that ALL current controls are set to zero (turned fully counterclockwise).

**NOTE:** Make sure the Teslometer is on at least one hour before taking either measurement to ensure it is stable.

2. Remove LOTO, connect the input power to the S/C Shim Coil Power Supply, and turn on (I) all Shim Coil Switch Heaters for five (5) minutes to dump any residual shim currents.
3. Turn the Switch Heaters off (O). Allow three to five minutes for the Switch Heaters to go persistent before continuing.
4. Disconnect the Heater Cable P703 from the Shim Coil Power Supply.
5. Turn off the Shim Coil Power Supply.



### **WARNING**

**ELECTRICAL SHOCK HAZARD!**  
CONTACT WITH CONNECTORS LEADING TO AN ENERGIZED POWER SUPPLY CAN CAUSE ELECTRICAL SHOCK.  
FOLLOW LOTO PROCEDURES WHILE MAKING ELECTRICAL CONNECTIONS TO THE SHIM POWER SUPPLY.

6. Disconnect and LOTO the input power to the Shim Coil Power Supply. Check that no voltage is present by using a DVM or equivalent measuring device.
7. Measure and record the magnetic field (center frequency) as soon after ramping as possible. This is Frequency 1.



### **NOTICE**

Do NOT move the Probe between the first and second measurements.  
Do NOT move or rearrange any articles or equipment in or near the Exam Room during the test.

8. After 12 hours, repeat [Step 2](#) through [Step 7](#). Measure and record this as Frequency 2.
9. Check that the magnet has stabilized (0.1 ppm/hr. maximum drift) by using the following formula:

$$\text{Drift Rate (ppm/hr)} = \frac{(\text{Freq 1} - \text{Freq 2}) \times 10^6}{(\text{Freq 1}) \times N}$$

$$\text{Drift Rate (Hz/hr)} = \frac{(\text{Freq 1} - \text{Freq 2})}{N}$$

Where:

Freq 1 = Initial Reading (Hz)

Freq 2 = Reading (Hz) after N Hours

N = Number Hours between Readings

17 Oct. 2006

#### 11.4.1.4 Magnet Pressure, Helium Level, and Coldhead S/C Shield

Shimming is affected by magnet pressure. Make sure the Magnet Monitor Pressure Controller is functioning properly and magnet pressure is steady at 3.9 to 4.1 psig before starting the shimming procedure.

1. On the Magnet Monitor:

- a. Check the magnet pressure. Make sure it is maintained between 3.9 to 4.1 psig when taking maps of the magnetic field.
  - b. Check that the magnet helium level is  $\geq 85\%$ . Fill the magnet in conformance with Liquid Helium Fill to make sure the helium level is  $\geq 85\%$  before continuing.
2. Turn the Cryocooler off for 10 minutes. (This quenches the S/C Shield around the Coldhead.)

**NOTE:** Only quench before the initial shimming and not between each shimming iteration.

#### 11.4.1.5 Field Camera Set-Up Using ShimTool



### NOTICE

Always use the most recent revision of GE Shimtool Software CD 5102783. The MR450 Shimming procedure stated here includes shimming of RB series LCC magnets.

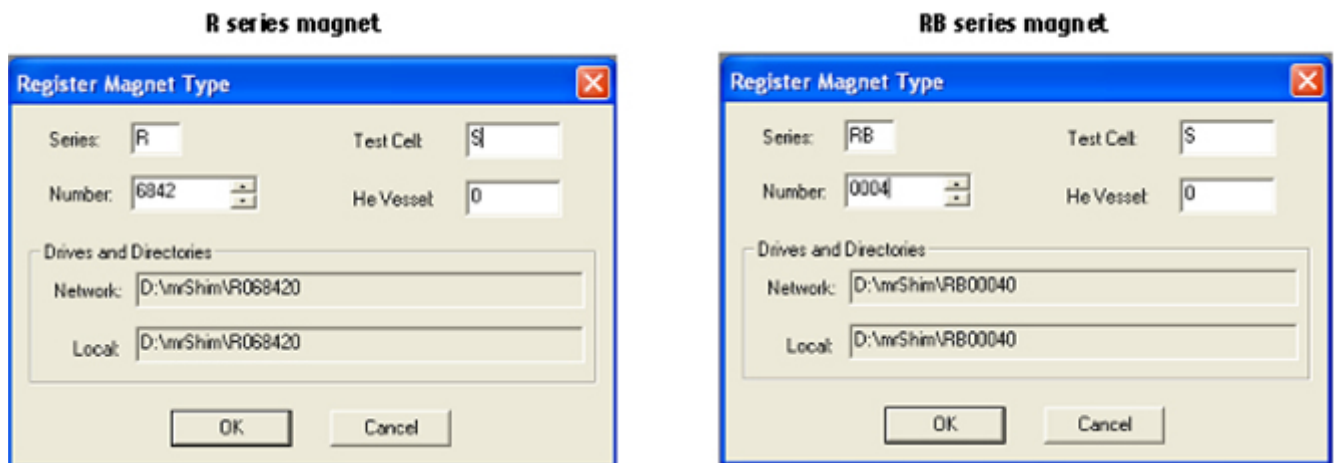
1. Connect the Shim Camera to the PC's COM port, and make sure all connections are secure.
2. Install the Mapping Fixture's Probe Array (if not done already) in conformance with [Preparations for Field Measurement](#).
3. If the most recent version of GE Shimtool Software CD 5102783 is **not** installed on the computer being used:
  - a. Insert the CD into the computer, which automatically launches the installation program. When asked, select the operating system (either Windows 2000 or Windows XP) for the computer in use. The CD should automatically install all ShimTool files at D: \MRshim (Windows XP) or C: \MRshim (Windows 2000).
  - b. Create a subdirectory for this magnet's .map, .cur, and .shm files inside the D: \MRshim (Windows XP) or C: \MRshim (Windows 2000) directory where ShimTool was installed.
    - For R Series magnets, name the subdirectory R0####0 (substituting the magnet's serial number for the ####).
    - For RB Series magnets, name the directory RB####0.

Add extra zeros as necessary to make sure the subdirectory name is seven characters long. The subdirectory for R series magnet 22 should be R000220.
4. Double-click the CoronaShim icon or the ShimTool\*.exe listing in the ShimTool directory in Windows Explorer.
5. When the *Register Magnet Type* screen shown below displays:

- a. Check that the entry next to **Series**: matches the **R** or **RB** of the magnet to be shimmed.
- b. Type **W** into the box next to **Test Cell**:. (Use **W** for all shimming done at a customer's site.)
- c. Adjust the numerical value next to **Number**: to the serial number of the magnet to be shimmed.
- d. Click [OK].

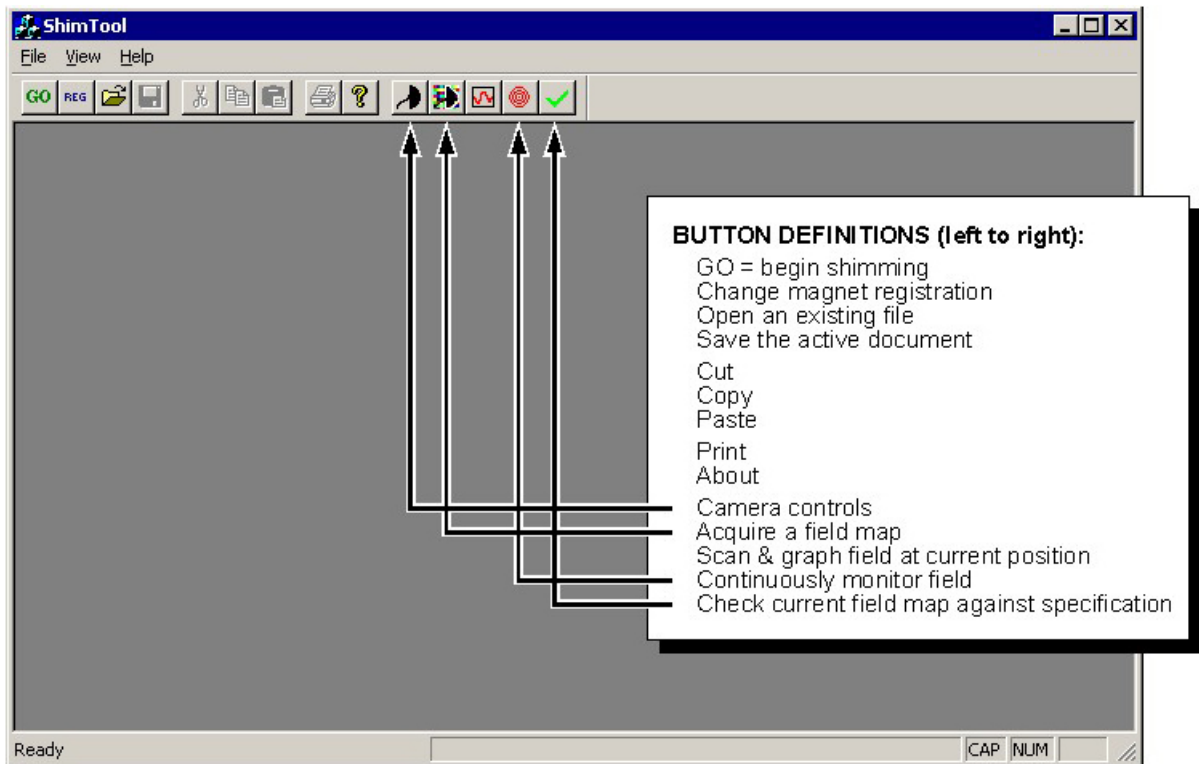
**NOTE:** If a subdirectory for storing all .map, .cur, and .shm files was not manually created earlier, ShimTool creates one automatically at **D:\MRshim\R0####0** for R Series magnets or at **D:\MRshim\RB####0** for RB Series magnets. (### represents the serial number selected.)  
Windows XP installations use the D: drive; Windows 2000 uses the C: drive.

**Illustration 3-154: Register Magnet Type Screen**



6. Select [Camera Controls] (fifth button from right) when the *ShimTool*/main screen shown below displays.

Illustration 3-155: ShimTool Main Screen Button Definitions

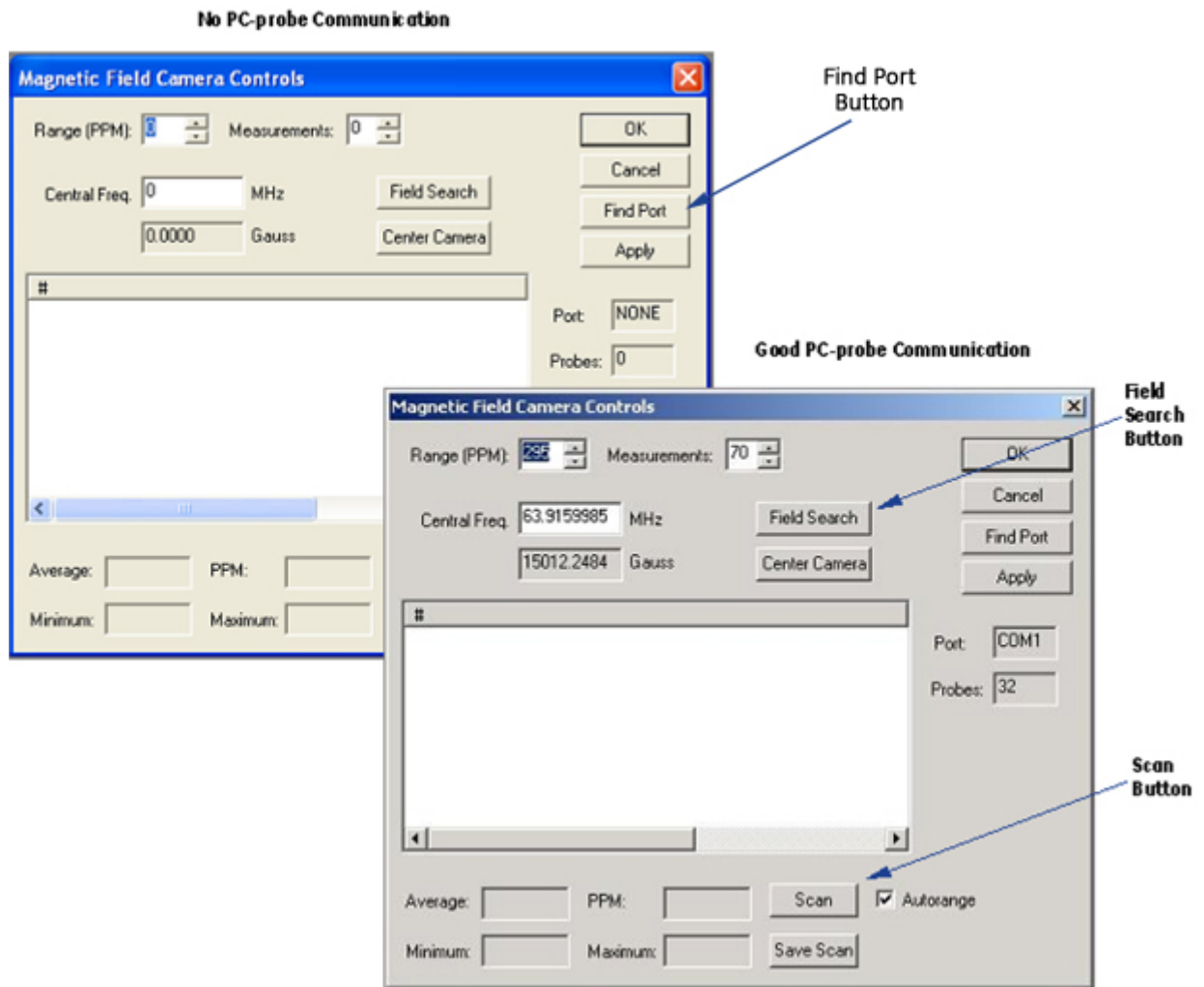


7. Make sure the *Magnetic Field Camera Controls* window shown below displays a **Center Freq:** of **63.8 MHz** (rounded).

If the program did not find a camera attached to the PC's COM port (**NONE** displays in the **Port:** field), proceed as follows:

- Check that power to the camera is turned on and all connections between the Probe Array, Camera, and PC are secure.
- Click the [Find Port] button to reestablish Camera-PC communications.
- Exit any programs that Windows Task Manager (accessed by pressing <Ctrl> + <Alt> + <Delete>) is running on the PC besides those that ShimTool and Windows Explorer indicate.

Illustration 3-156: Magnet Field Camera Controls: PC-Probe Communication



8. Choose an option on the *Magnetic Field Camera Controls* screen shown above:
  - a. Click the [Field Search] button to automatically find the center frequency of the magnet, and update the **Center Freq:** field.

Illustration 3-157: Magnet Field Camera Controls (After Choosing Field Search Button)

**Magnetic Field Camera Controls**

Range (PPM): 295 Measurements: 70

Central Freq: 63.9159999 MHz

15012.2484 Gauss

Field Search

Center Camera

| #  | Freq [MHz] | Field [G]   | St... | R... |
|----|------------|-------------|-------|------|
| 00 | 63.9081358 | 15010.40161 | 1.80  | 070  |
| 01 | 63.9082145 | 15010.42010 | 1.80  | 070  |
| 02 | 63.9097144 | 15010.77239 | 2.40  | 070  |
| 03 | 63.9103800 | 15010.92872 | 1.80  | 070  |
| 04 | 63.9107592 | 15011.01778 | 1.80  | 070  |
| 05 | 63.9109203 | 15011.05562 | 1.20  | 070  |
| 06 | 63.9111182 | 15011.10210 | 2.40  | 070  |
| 07 | 63.9114065 | 15011.16982 | 1.80  | 070  |
| 08 | 63.9115240 | 15011.19741 | 1.20  | 070  |
| 09 | 63.9111625 | 15011.11251 | 1.20  | 070  |
| 10 | 63.9104256 | 15010.93943 | 1.80  | 070  |

Average: 63.913855 PPM: 246.086

Minimum: 63.908136 Maximum: 63.923864

Port: COM1

Probes: 32

Scan

Save Scan

OK

Cancel

Find Port

Apply

☒ Autorange

An initial listing from left to right in the scrolling text box:

- Probe number: #,
- Frequency in MHz: **Freq (MHz)**,
- Magnetic field strength in Gauss: **Field (G)**,
- Standard deviation in Gauss, and
- Measurement count readings from each probe.

**NOTE:** The measurement counts should all read close to the number next to **Measurements**: at the top of the screen.

- Click the [Scan] button to verify the probe function by updating the scrolling text box.
- Click [OK] to finish the camera set-up and close the *Magnetic Field Camera Controls* screen.
  - Proceed to the next section, Zeroing Currents.

### 11.4.2 Zeroing Currents



#### **⚠ DANGER**

POTENTIAL FATAL INJURY!

BEFORE STARTING THE SHIMMING PROCEDURE:

MAKE SURE TO REVIEW AND FULLY UNDERSTAND ALL S/C MAGNET PORTIONS OF [CHAPTER 1, MR MAGNET - SAFETY REQUIREMENTS](#).



#### **⚠ WARNING**

ASPHYXIATION HAZARD!

THE SHIMMING PROCEDURE GENERATES ODORLESS, COLORLESS HELIUM GAS THAT DISPLACES OXYGEN IN THE AIR.

MAKE SURE MAGNET ROOM VENT EXHAUST FAN IS ON BEFORE STARTING THE SHIMMING PROCEDURE.



#### **⚠ CAUTION**

Potential Magnet Quench!

The magnet may quench if the Nupro Valve on the Shim Lead is closed or the Shim Lead is not cold.

Make sure the Nupro Valve is fully open, and the valve is frosted before slowly engaging the Shim Lead. Do NOT remove the Shim Lead's Orifice Cap that controls helium gas flow. Always engage/disengage the Shim Lead in conformance with Shim Lead Engagement and Disengagement.



#### **NOTICE**

Ensure that the shim power supply is calibrated in conformance with the vendor's manual and the shim heater currents are set as stated below before use:

- Transverse 1 Shim Heater Current: 610 mA,  $\pm 5$  mA
- Transverse 2 Shim Heater Current: 610 mA,  $\pm 5$  mA
- Axial Shim Heater Current: 710 mA,  $\pm 10$  mA



#### **NOTICE**

Before turning on the shim coil power supply and applying shim currents, ensure the following conditions exist to prevent irreparable damage to the vapor-cooled Shim Leads:

- Orifice on the Shim Lead Nupro Vent Valve is in place.
- Shim Lead is frosting and venting.
- Shim Power Supply current controls are set to zero (fully CCW).



## NOTICE

The Recommended Total Current values found on the Acceptance Test Report (ATR) that shipped with the magnet must be entered into the S/C Correction Coils before performing field mapping and any further S/C or passive shimming.

The coil groups are:

- Transverse 1 Coils: T1-1 to T1-6
- Transverse 2 Coils: T2-1 to T2-6
- Axial Coils: AX1 to AX6

Each coil group is connected to one heater circuit on the Shim Power Supply. If the only current change was in the Axial 2 Coil, turning the Axial Switch Heater on would dump the currents in the Axial Coils (AX1 to AX6), and the appropriate currents for each Axial Coil need to be entered.

Always check the ATR for any possible magnet-specific current settings.

1. After ensuring that vapor is escaping from the Shim Lead and the lead is frosted, turn on the power supply.



## NOTICE

Do NOT move articles or equipment in the Magnet Room while adjusting shim currents.

2. Repeat the following sub-steps to input the ATR shim currents for each shim group in the recommended sequence: Transverse 1 (T1), Transverse 2 (T2), and finally Axial in the table below.

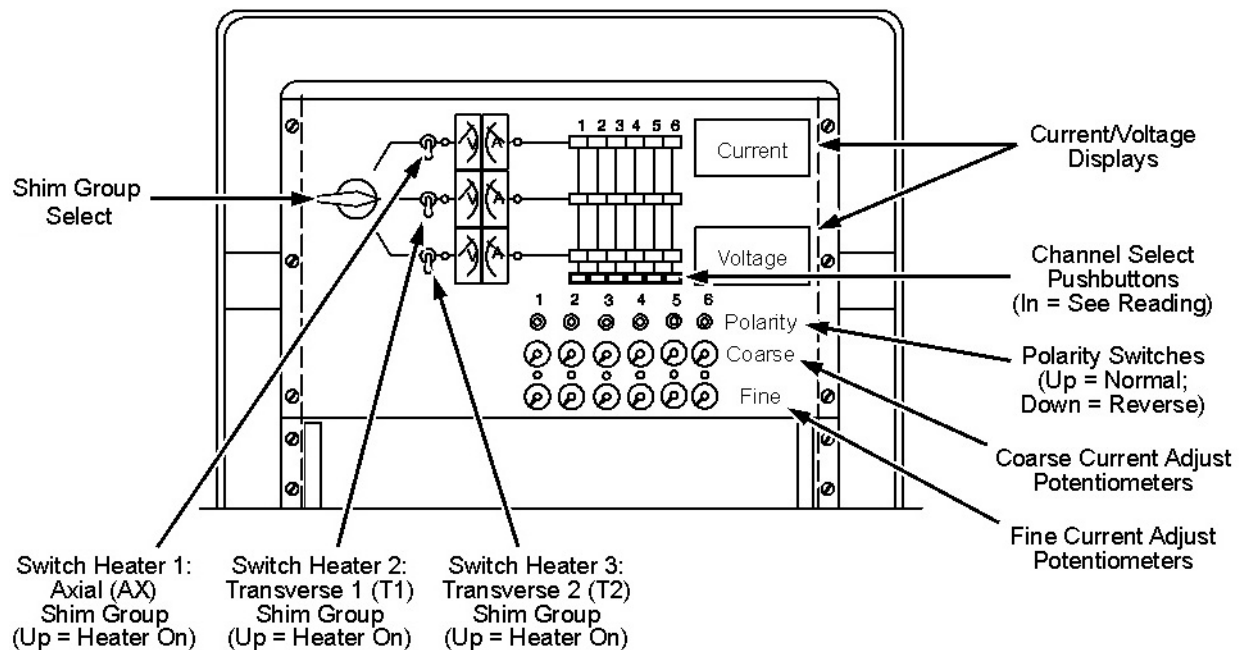
a.

Table 3-20: S/C Shim Groups

| Course/Fine Adjust Potentiometer & Polarity Light/Switch | Transverse 1 Shim Group          | Transverse 2 Shim Group | Axial Shim Group      |
|----------------------------------------------------------|----------------------------------|-------------------------|-----------------------|
|                                                          | Shim Group Select Switch Setting |                         |                       |
|                                                          | Middle                           | Bottom                  | Top                   |
|                                                          | Coil (Circuit)                   | Coil (Circuit)          | Coil (Circuit)        |
| 1                                                        | T1-1                             | T2-1                    | AX1 (Z <sup>1</sup> ) |
| 2                                                        | T1-2                             | T2-2                    | AX2 (Z <sup>2</sup> ) |
| 3                                                        | T1-3                             | T2-3                    | AX3 (Z <sup>3</sup> ) |
| 4                                                        | T1-4                             | T2-4                    | AX4 (Z <sup>4</sup> ) |
| 5                                                        | T1-5                             | T2-5                    | AX5 (Z <sup>5</sup> ) |
| 6                                                        | T1-6                             | T2-6                    | AX6 (Z <sup>6</sup> ) |

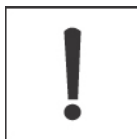
- b. Set the Shim Group Select Switch for the appropriate group: T1, T2 or Axial.

Illustration 3-158: Shim Power Supply Controls (Front Panel)



- c. Set the Course/Fine Current Adjust Potentiometers on the Shim Power Supply to zero (0) Amps (fully CCW).

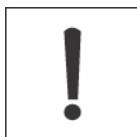
**NOTE:** The polarity relays do not change polarity if there is any current flowing through their respective channels. Reduce the current to zero (0) before the relays change to the position indicated by the external switches on the power supply.



## NOTICE

Magnet damage is possible if heater currents are not set correctly.

- d. Turn on the selected Switch Heater group (T1, T2, or Axial), and wait 30 seconds for the switches to go resistive.



## NOTICE

Failure to accurately set the Power Supply currents and polarity can greatly increase the number of plotting iterations (i.e., time to shim the magnet).

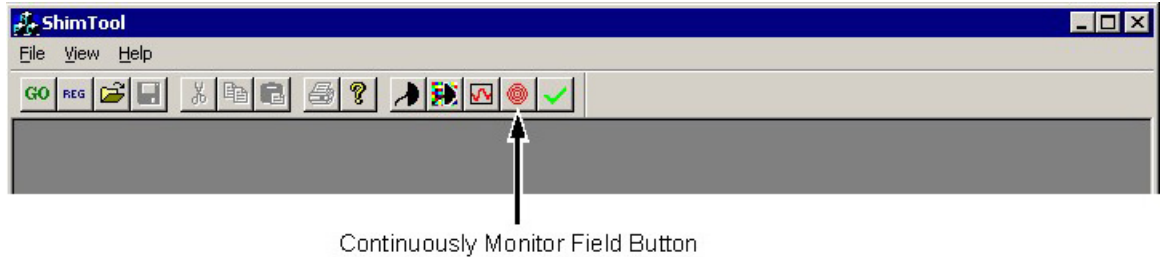
- e. Turn on all Switch Heater groups (T1, T2, and Axial) with current output set to zero (0), wait five minutes, then turn them off.
- f. After waiting five (5) minutes, gradually turn the power supply Current Adjust Potentiometers down to zero (0) Amps.

### 3. (For Axial Currents Only)

**(For Axial Currents Only)** Monitor the magnetic field drift until it stabilizes (i.e., <2 Hz change in total magnetic field strength over a two-minute period).

- a. Click [Continuously Monitor Field] (second button from right) on the *ShimTool* main screen.

**Illustration 3-159: Continuously Monitor Field Button on ShimTool Main Screen**



- b. Ensure the probe array is in the 30° position (Slice/Arc 2 in the table below), then select **Probe 22**.

**Table 3-21: Field Camera Mapping Slices (Positions)**

| Slice/Arc <sup>a</sup><br>(Position) | Degrees            | Slice/Arc <sup>a</sup><br>(Position) | Degrees         | Slice/Arc <sup>a</sup><br>(Position) | Degrees          | Slice/Arc <sup>a</sup><br>(Position) | Degrees          |
|--------------------------------------|--------------------|--------------------------------------|-----------------|--------------------------------------|------------------|--------------------------------------|------------------|
| 0                                    | 0/360 <sup>b</sup> | 6                                    | 90 <sup>b</sup> | 12                                   | 180 <sup>b</sup> | 18                                   | 270 <sup>b</sup> |
| 1                                    | 15                 | 7                                    | 105             | 13                                   | 195              | 19                                   | 285              |
| 2                                    | 30                 | 8                                    | 120             | 14                                   | 210              | 20                                   | 300              |
| 3                                    | 45                 | 9                                    | 135             | 15                                   | 225              | 21                                   | 315              |
| 4                                    | 60                 | 10                                   | 150             | 16                                   | 240              | 22                                   | 330              |
| 5                                    | 75                 | 11                                   | 165             | 17                                   | 255              | 23                                   | 345              |

<sup>a</sup> The Slice/Arc number shown here and in ShimTool corresponds with the number stamped on the mapping fixture's positioning wheel.  
<sup>b</sup> The remote box flashes slower at these positions.

- c. Click the [Start] button on the screen that displays to begin measuring field strength and drift. Make a note of the Starting Frequency for use in the next step.
- d. Wait two (2) minutes, click the [Stop] button, and record the change between the starting and present frequencies.
  - If the change is <2 Hz, continue to the next step.
  - If the change is ≥2 Hz, repeat [Step 3.c](#) and [Step 3.d](#).
4. Once the magnetic field is stable and the Axial settling time elapses, turn off the Axial Switch Heater and allow it to cool for five (5) minutes.
5. Slowly turn all the Course/Fine Current Adjust Potentiometers on the Shim Power Supply down to zero amps (fully CCW).

6. Close the Shim Lead's Nupro Vent Valve.
7. Proceed to the next section, Running ShimTool.

### 11.4.3 Running ShimTool

#### 11.4.3.1 Field Map Acquisition



#### NOTICE

The magnet's center frequency must be stable in conformance with [Section 11.4.1.3](#), Field Stability Check, before any alteration to the magnet's S/C or passive shims are made.



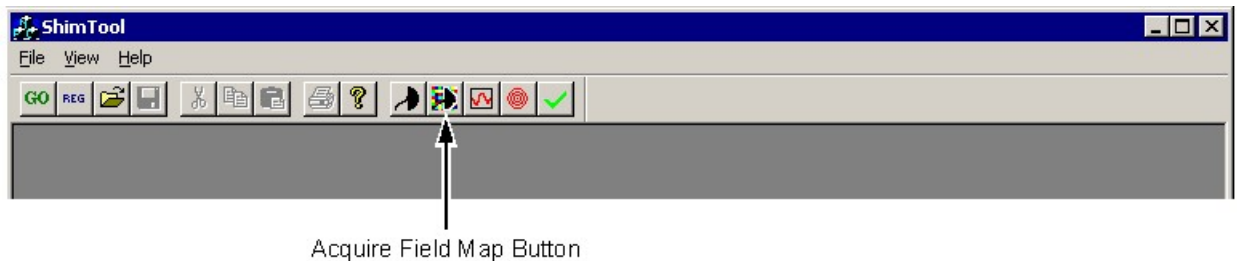
#### NOTICE

Moving articles or equipment in the Magnet Room may affect readings of magnetic field strength.

**NOTE:** Locate the Mapping Fixture and Shim Camera at the magnet's approximate center in conformance with [Preparations for Field Measurement](#) before centering the Probe Array using ShimTool.

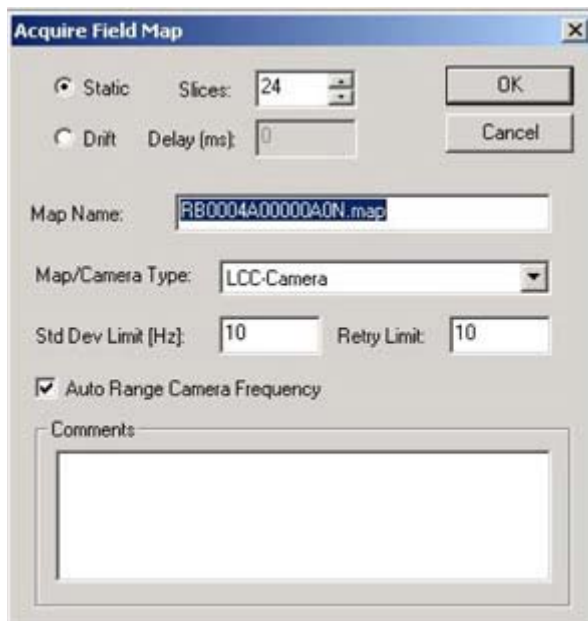
1. Click [Acquire Field Map] (fourth button from right) on the *ShimTool* main screen.

Illustration 3-160: Acquire Field Map Button on ShimTool Main Screen



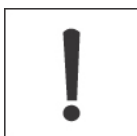
2. Click the [Static] radio button on the *Acquire Field Map* screen, then select **LCC** from the **Map / Camera Type**: drop-down menu.

Illustration 3-161: Static Button on Acquire Field Map Screen



**NOTE:** Do not change the default number for **Slices:** and the filename for **Map Name:**.

3. Make sure the curved part of the field camera's probe array faces directly up (0° position).

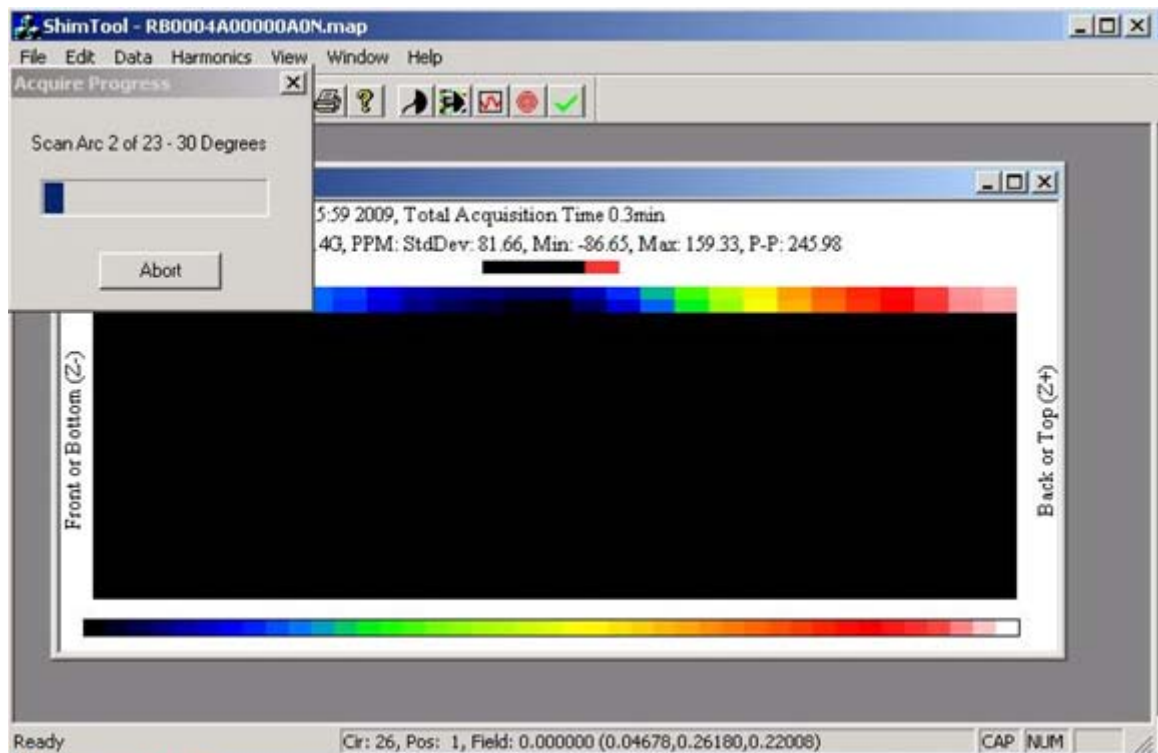


## NOTICE

Moving any articles or equipment in or near the Magnet Room may affect the field map. Do NOT allow nonessential conversation/distractions during the map acquisition.

4. If an error occurs during Field Map Acquisition:
  - a. Click the [Abort] button on the *Acquire Progress* window.

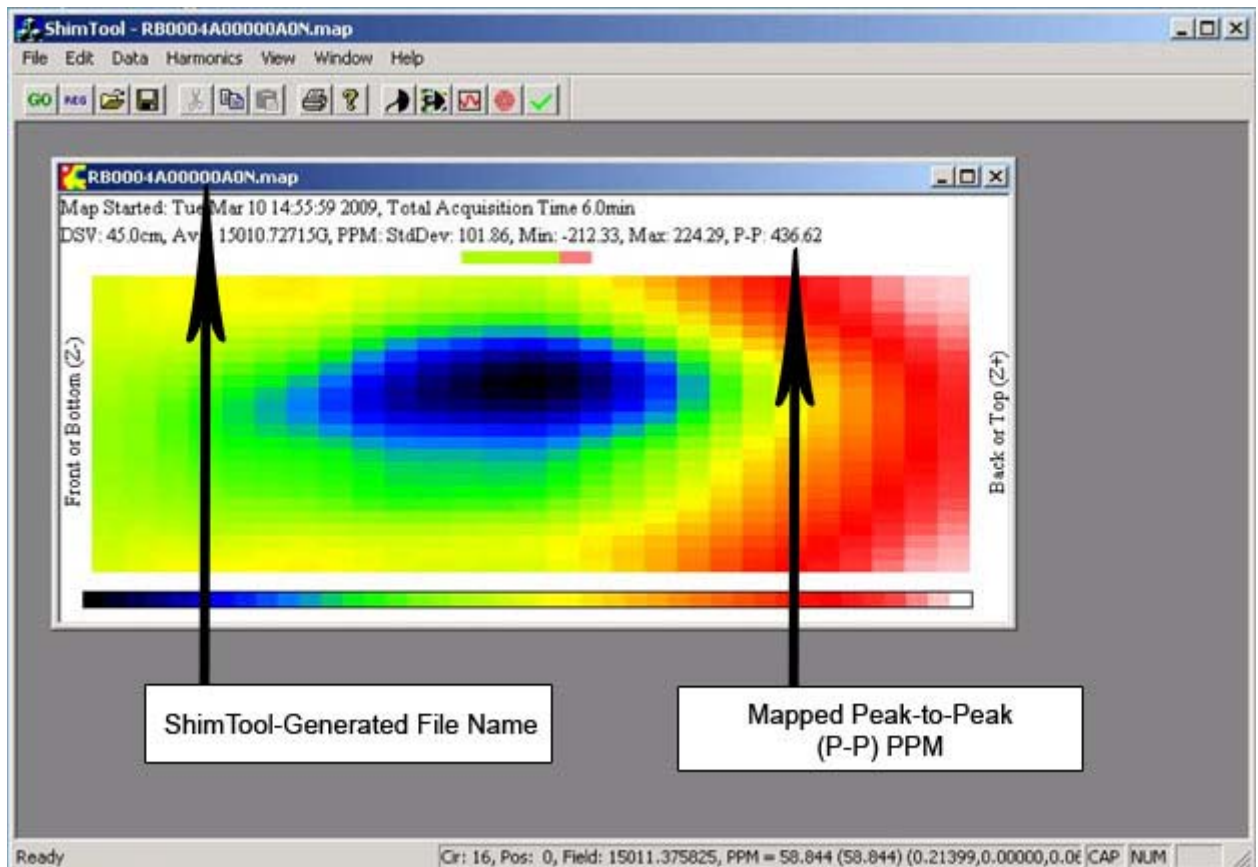
Illustration 3-162: Field Map Being Acquired



- b. Press the field camera's remote box switch one more time.
  - c. Retake the map starting with [Step 1](#).
5. Click [OK] to begin mapping the magnet's magnetic field.

**NOTE:** The Field Map screen displays with the *Acquire Progress* window overlaid as shown in [Illustration 3-162](#). The field camera's remote box switch should start to repeatedly flash.
6. Press the field camera's remote box switch when it starts to flash repeatedly. The light stops flashing and remains illuminated while the system acquires one slice of the field map. After the last slice, the light goes off.
7. When the camera's remote box switch starts to flash repeatedly again, slowly and smoothly rotate the camera one position (15°) clockwise.
8. Repeat the above two steps to take 24 map slices: Slice 0 at 0°/360° through Slice 23 at 345°. (See [Table 3-21](#) for a summary of the slices.)
  - The camera's remote box switch flashes slower at positions 0 (0°/360°), 6 (90°), 12 (180°) and 18 (270°).
  - After taking all 24 slices, the finished map displays on the ShimTool Field Map screen shown below.

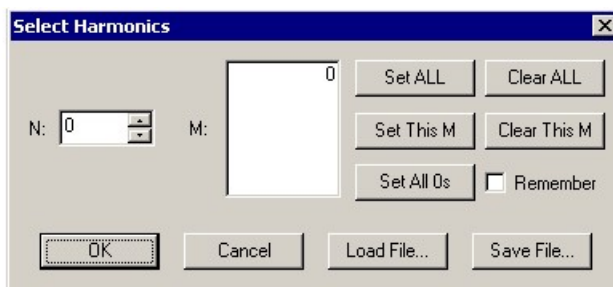
Illustration 3-163: Sample First Field Map Screen for Axial Centering



**NOTE:** The ShimTool automatically assigns a sequential alphabetic filename to both the displayed map and the .map file it creates.

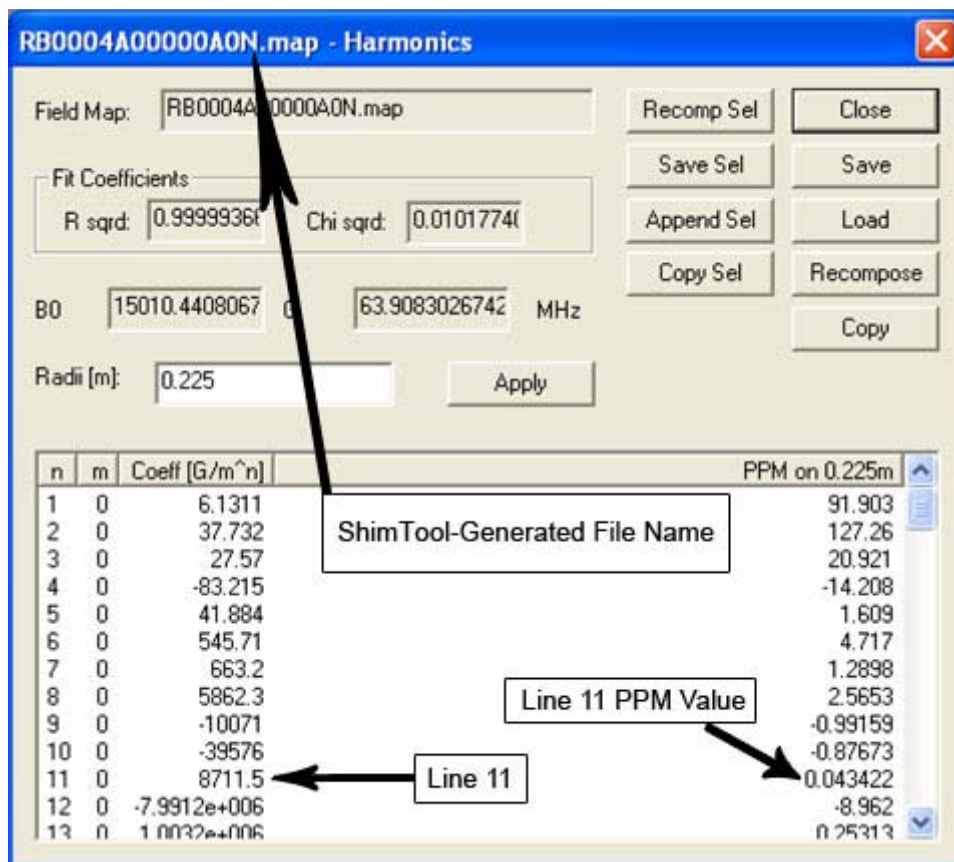
- For Magnet **Series:** and **Number:** entered on the *Register Magnet Type* screen ([Illustration 3-154](#)), R0 + 0009 becomes R00009 and RB + 0002 becomes RB0002.
  - A sequential alphabetic character A, B, ... C becomes: 0000A, 0000B, ... 0000C.
  - Use R when shimming at a customer site for **Test Cell:** entered on the *Register Magnet Type* screen.
9. Record the mapped Peak-to-Peak (P-P) ppm value displayed on the Field Map screen into the Shimming Process Data on the same line as the map name.
  10. If this is the first Field Map taken since the mapping fixture was installed or moved in any substantial way, use this map to axially center the Probe Array:
    - a. Select the *Harmonics* menu (fourth item from the left, top of [Illustration 3-163](#)), and click **Decompose (F6)**.
    - b. Click [OK] when the *Select Harmonics* screen displays.

Illustration 3-164: Select Harmonics Screen



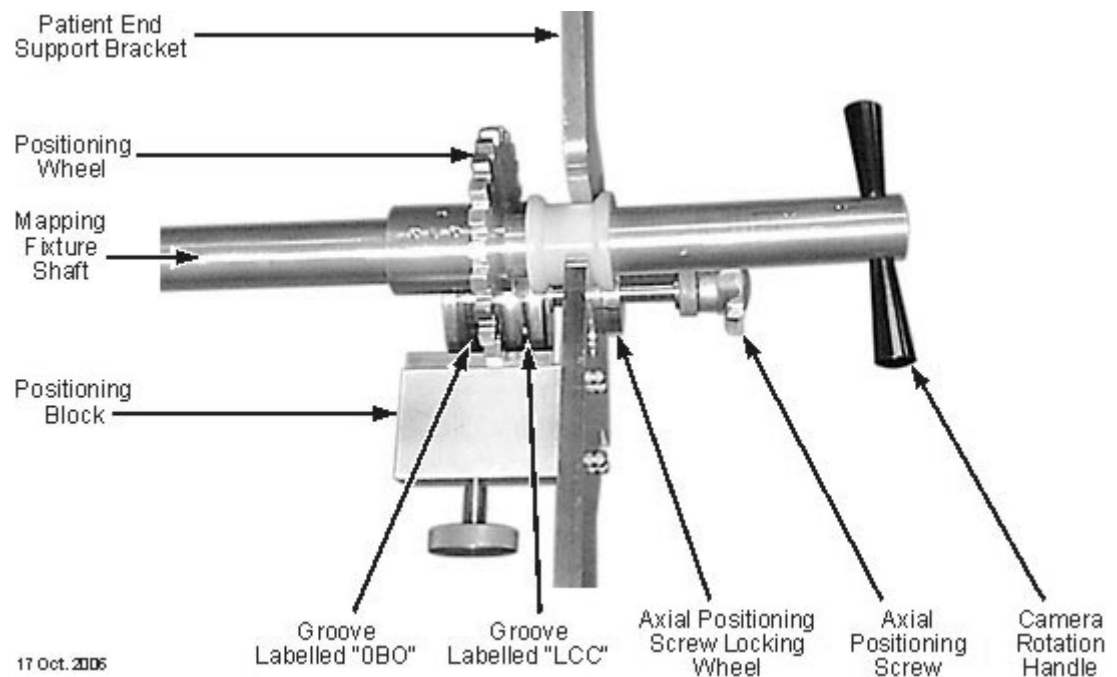
- c. Look at the PPM value displayed on the right side of line 11 when the screen shown below displays.

Illustration 3-165: Field Map Harmonics



- d. If the PPM value on Line 11 is  $>-0.8$  ppm and  $<+0.8$  ppm, the Probe Array is centered axially. Continue with [Step 11](#), and use this centered map to run ShimTool, set for S/C Shimming.
- e. Verify that the Positioning Wheel on the Mapping Fixture is in the groove labeled LCC.

Illustration 3-166: Mapping Fixture 5266042 (Side View)



- f. Divide the PPM value obtained in [Step 10.c](#) by 0.75. (See the example below.)

|                      |   |                                                                                                           |
|----------------------|---|-----------------------------------------------------------------------------------------------------------|
| PPM Value on Line 11 | = | Number of Turns<br>of Mapping Fixture's Axial Positioning Screw<br>Required to Axially Center Probe Array |
| 0.75                 |   |                                                                                                           |

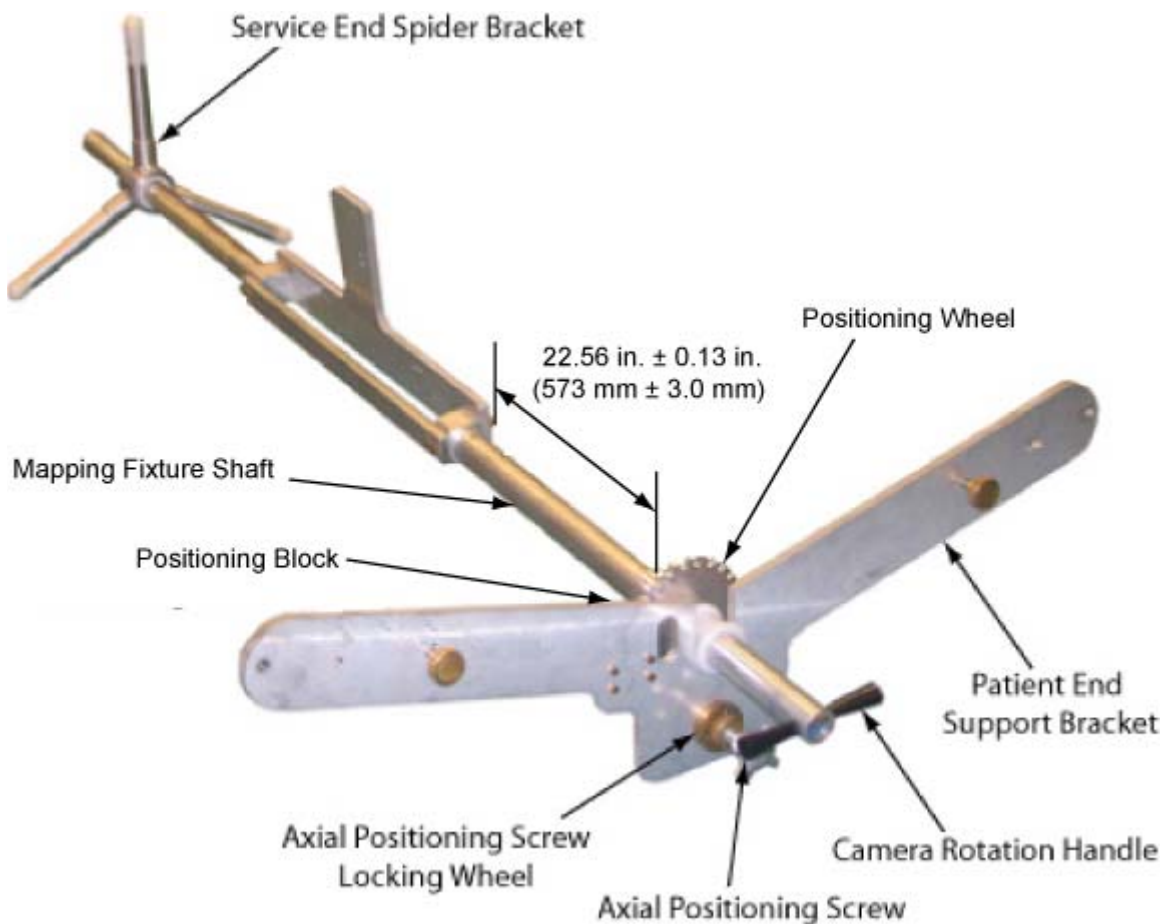
- g. Loosen the Axial Positioning Screw Locking Wheel, carefully turn the Axial Positioning Screw the number of full turns calculated with the above formula.

**NOTE:** Turning in increments smaller than a quarter turn is not necessary.

**On normal polarity magnets:**

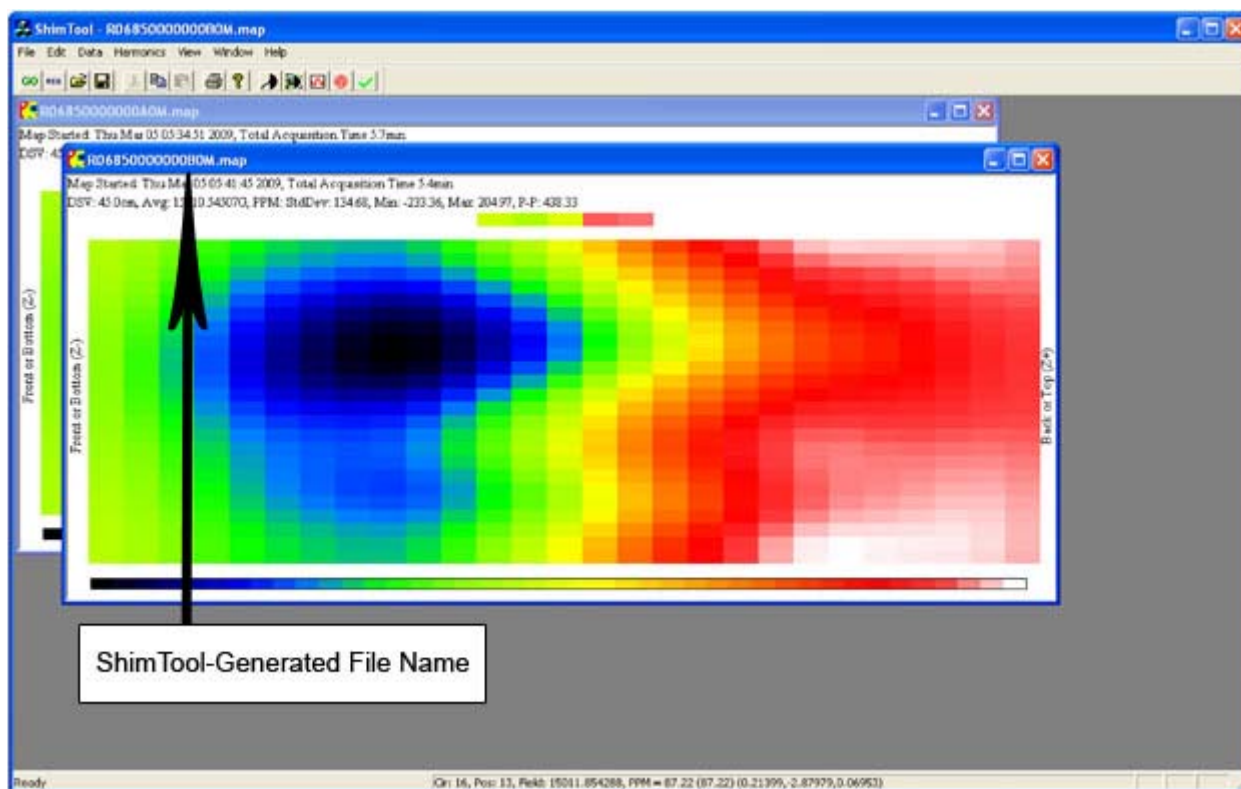
- For a positive (+) PPM value: Turn the Axial Positioning Screw clockwise.
- For a negative (-) PPM value: Turn the Axial Positioning Screw counterclockwise.

Illustration 3-167: Mapping Fixture 5266042 with Spider Bracket (Front View)



- h. Tighten the Locking Wheel.
  - i. Record the ShimTool generated **.map** filename into the Shimming Process Data in the upper section, Maps for Probe Centering.
  - j. Repeat [Step 1](#) through [Step 10.d](#) to verify axial centering of the Probe Array.
11. Record the ShimTool generated **.map** filename (shown in [Illustration 3-168](#)):
- Map names used for hybrid shimming (passive and S/C) into the Shimming Process Data, Passive Shimming Data, and SC Shimming Data.
  - Map names used when running ShimTool for only S/C into *Shimming Process Data* and *SC Shimming Data*.

**Illustration 3-168: Sample Field Map Screen for Shimming**

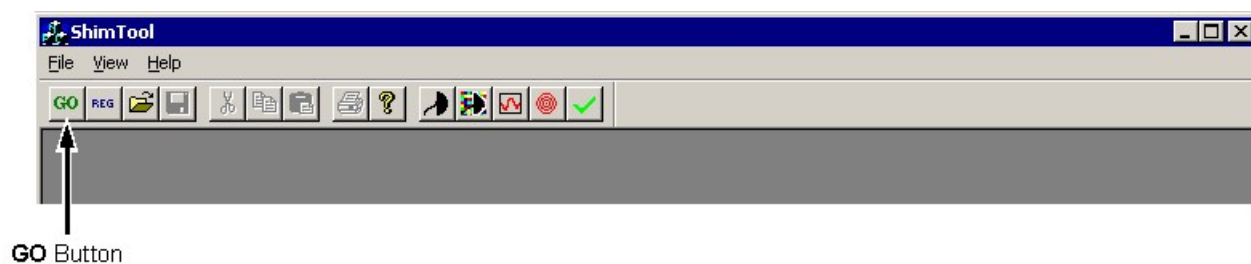


12. Close the ShimTool Harmonics window(s).

### 11.4.3.2 Acquiring a Shim File

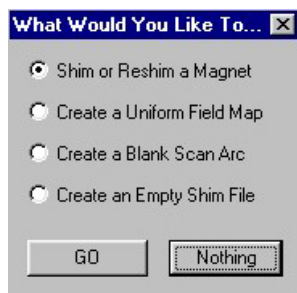
1. Click [GO] (button farthest left) on the *ShimTool* main screen.

**Illustration 3-169: GO Button on ShimTool Main Screen**



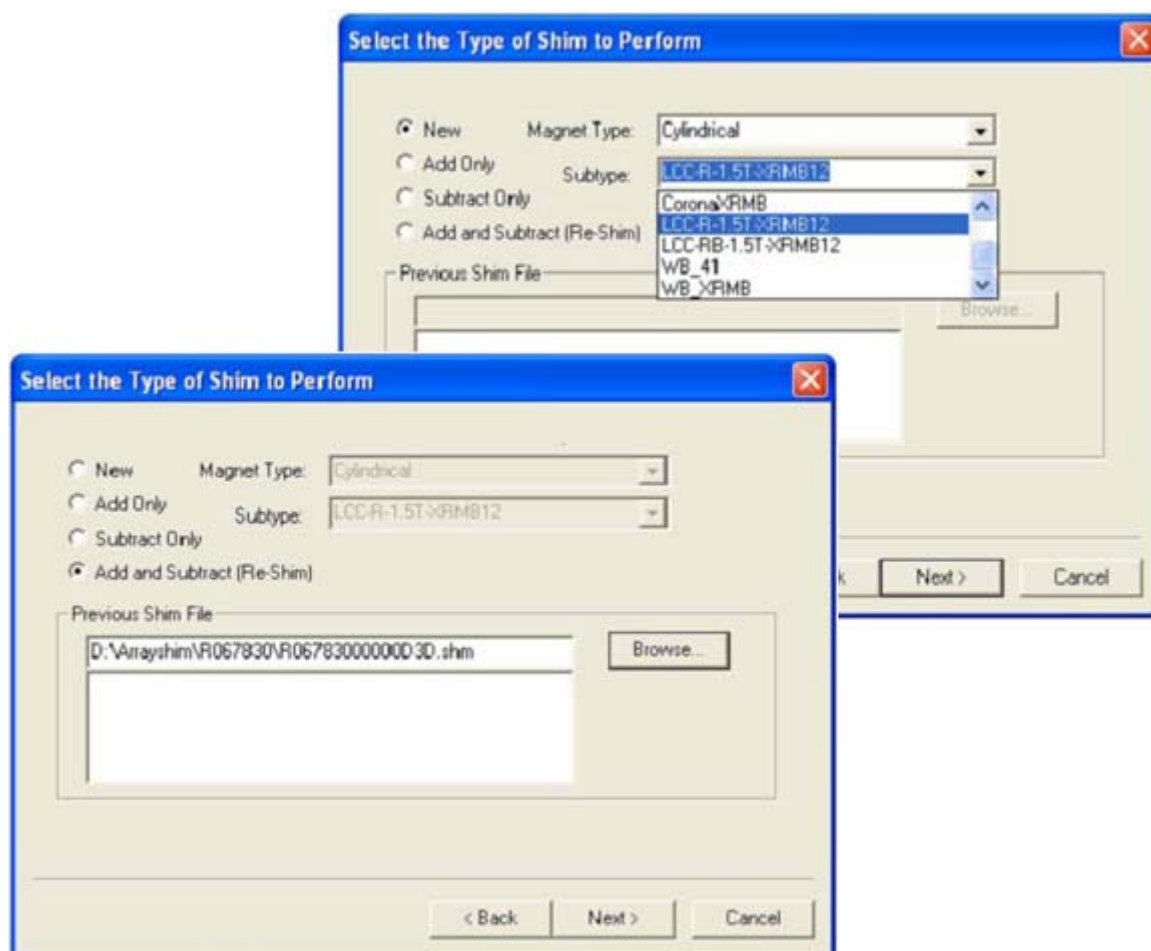
- Click the [Shim or Reshim a Magnet] radio button, and click [GO] when the *What Would You Like To...* screen displays.

Illustration 3-170: What Would You Like to Do? Screen



3. When the *Select the Type of Shim to Perform* screen displays:
  - a. Check that the **Magnet Type**: field reads **Cylindrical**.

Illustration 3-171: Select the Type of Shim to Perform Screen



- b. Select the correct **Subtype**: stated in the table below.

Table 3-22: Magnet Subtype and Calibration File Choices

| Series | Positions | Correct Choices    |                          |                         |
|--------|-----------|--------------------|--------------------------|-------------------------|
|        |           | Magnet Subtype     | Passive Calibration File | Active Calibration File |
| R      | 45        | LCC-R-1.5T-XRMB12  | XRMB_0_25mm.cal          | RS393SC.cal             |
| RB     | 45        | LCC-RB-1.5T-XRMB12 | XRMB_0_25mm.cal          | RB_SC.cal               |

- c. Select the [New] radio button. (ShimTool can only run with this radio button selected at the factory.)



## NOTICE

**Avoid using incorrect filenames.**

Keep track of filenames and iterations used (and what they were used for) by entering this information into the Shimming Process Data (master log of the shimming process) and the Data Sheets affected by Superconducting Shimming (SC Shimming Data), and by hybrid shimming (Passive Shimming Data and *SC Shimming Data*) so that accurate filenames can be used at any later time.

**NOTE:** If the output file is not located as stated in the table below and the full path to the correct shim file cannot be recalled, click [Browse] (shown in [Illustration 3-171](#)), locate the correct file, select it, and open the file.

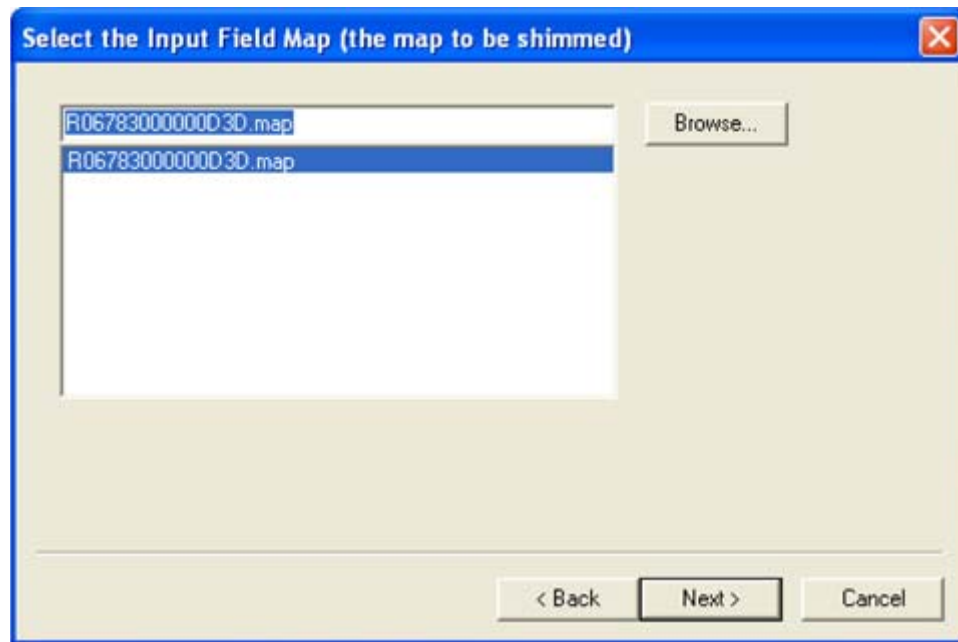
Table 3-23: Directory Path, Filenames and File Format

| Magnet                                                                                                                                                                                                                                                                                                                         | Series | File Type | Directory Path to File        |                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------|-------------------------------|-------------------------------|
|                                                                                                                                                                                                                                                                                                                                |        |           | Windows 2000                  | Windows XP                    |
| 1.5T LCC                                                                                                                                                                                                                                                                                                                       | R      | SC        | C:\MRshim\R0####0\R0####0.cur | D:\MRshim\R0####0\R0####0.cur |
|                                                                                                                                                                                                                                                                                                                                |        | Passive   | C:\MRshim\R0####0\R0####0.shm | D:\MRshim\R0####0\R0####0.shm |
|                                                                                                                                                                                                                                                                                                                                |        | Map       | C:\MRshim\R0####0\R0####0.map | D:\MRshim\R0####0\R0####0.map |
|                                                                                                                                                                                                                                                                                                                                | RB     | SC        | C:\MRshim\RB####0\RB####0.cur | D:\MRshim\RB####0\RB####0.cur |
|                                                                                                                                                                                                                                                                                                                                |        | Passive   | C:\MRshim\RB####0\RB####0.shm | D:\MRshim\RB####0\RB####0.shm |
|                                                                                                                                                                                                                                                                                                                                |        | Map       | C:\MRshim\RB####0\RB####0.map | D:\MRshim\RB####0\RB####0.map |
| Format for Directory Path: Drive letter + : \MRshim\ + seven character (maximum) subdirectory name + \ + filename.<br>Format for Seven-Character Subdirectory Name: Series number (box beside <b>Series</b> : + 0 if required) + Magnet number (box beside <b>Number</b> : ) + Helium vessel number (assigned at the factory). |        |           |                               |                               |

- Ensure that all information on the *Select the Type of Shim to Perform* screen ([Illustration 3-171](#)) is accurate, then click [Next >].
- When the *Select the Input Field Map* screen displays, accurately enter the filename of the last field map with .map extension acquired in [Section 11.4.3.1](#), Field Map Acquisition.

**NOTE:** If the output file is not located as stated in [Table 3-23](#) and the full path to the correct map file cannot be recalled, click [Browse], locate the correct file, select it, and open the file.

Illustration 3-172: Select the Input Field Map Screen

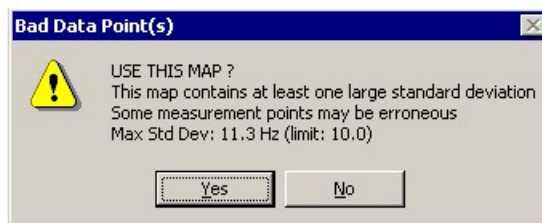


6. Click [Next >].

**NOTE:** The ShimTool looks at the data contained in the file specified.

- If that map file contains bad data, the alert shown below displays. (If the specified map file is good, this alert should not display.)
- Click [No], and retake the map in conformance with [Section 11.4.3.1](#), Field Map Acquisition. Ensure that all connections are good and the cable is not wrapping around the mapping fixture.

Illustration 3-173: Bad Data Point(s) Alert



7. Click the [Unified/Multi-Volume Shimming] radio button when the *Shim Target and Tolerance Controls* screen displays.
- a. Ensure the **Map Type**: field reads **LCC-4vol-393Unified**.

Illustration 3-174: Shim Target and Tolerance Controls

**Shim Target and Tolerance Controls**

☒ Unified/Multi-Volume Shimming    Map Type: LCC-4vol-393Unified

☒ Target (TOL) File: D:\Arrayshim\Shimtool25D\CALfiles\Rs393b.tol    Browse...

☐ Manual Target    PPM (Comma Separated)

☐ Auto-Tolerancing    ☐ Critical Targeting

PPM Threshold: 0    Release Amount [PPM]: 0

Change Limit: 0

**Tolerance File Name**

< Back    Next >    Cancel

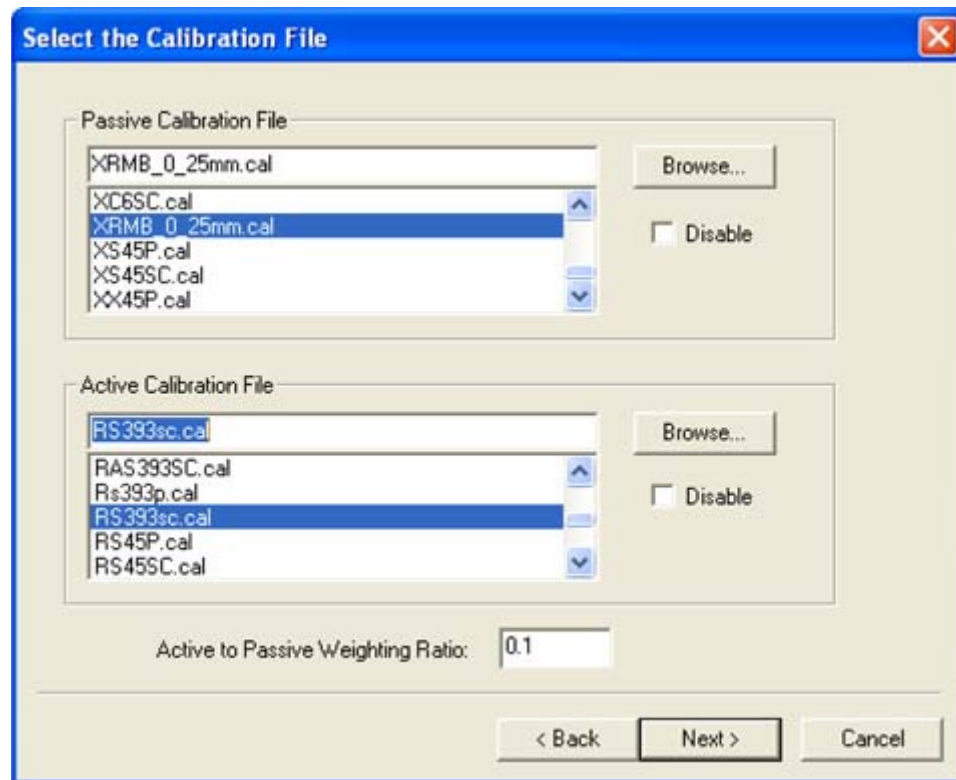
- b. Select the [Target (TOL) File] radio button.

If the ATR was produced using Tolerance Shimming, the path to **Rs393b.tol** displays next to **Target (TOL) File:**.

**NOTE:** The Windows XP path to the tolerance file should be: **D:\MRshim\ShimTool\Rs393b.tol**; Windows 2000 uses the C: drive. If the correct tolerance file is not located there and the full path to the correct location cannot be recalled, click [Browse], locate the correct file, select it, and open the file.

8. Click [Next >] to proceed to the *Select the Calibration File* screen.

Illustration 3-175: Select the Calibration File Screen



9. Select the **.cal** file from the Active Calibration File scrolling list (shown above) to match the number of shim positions stated in the ATR. (Table 3-22 shows the correct selection choices.)

**NOTE:** Shim tray positions are covered in Section 11.4.4, Passive Shim Installation.

10. Select the calibration files for both active and passive shims based on the magnet subtype. (See Illustration 3-175 and Table 3-22.) This applies to both R and RB Series magnets.

- For an R Series LCC, select **XRMB\_0\_25mm.cal** for the **Passive Calibration File** and **Rs393SC.cal** for the **Active Calibration File**.
- For an RB Series LCC, select **XRMB\_0\_25mm.cal** for the **Passive Calibration File** and **RB.SC.cal** for the **Active Calibration File**.

**NOTE:** If either the Active or Passive calibration files do not display in their respective scrolling lists, click [Browse], locate the correct file, select it, and open the file. The Windows XP path to the calibrations files should be **D:\MRshim\Shimtool\\*.cal** (where the \* represents the correct calibration file as stated in Table 3-22). Windows 2000 uses the C: drive.

11. If only performing one type of shimming (passive or active) during this iteration (shown in Illustration 3-148, Shimming Flowchart), click [Disable] (Illustration 3-175) next to the type of shimming **not** being done.

- Click [Disable] next to **Passive Shim Calibration File** if preparing to ONLY perform S/C or active Shimming ([Section 11.4.5](#), Superconducting Shimming).
- **Not recommended:** Click [Disable] next to **Active Shim Calibration File** if preparing to ONLY perform passive shimming ([Section 11.4.4](#), Passive Shim Installation). Normally, the **Active Shim Calibration File** is never disabled.

12. Click [Next >].



## NOTICE

ShimTool allocates the results of its calculations to all forms of shimming that are not disabled. Extra shimming iterations may result from not setting the disable boxes accurately to the type of shimming being done.

13. Click [Next >] without making changes on the *Disable Shims/Change Limits* screen when it displays.

Illustration 3-176: Disable Shims/Change Limits Screen

Disable Shims/Change Limits

Slot:

☒ Passive Shim  
Ring: 1

☐ Active Coil [Change Limits...](#)

Maximum Current/Thickness:  
12.00

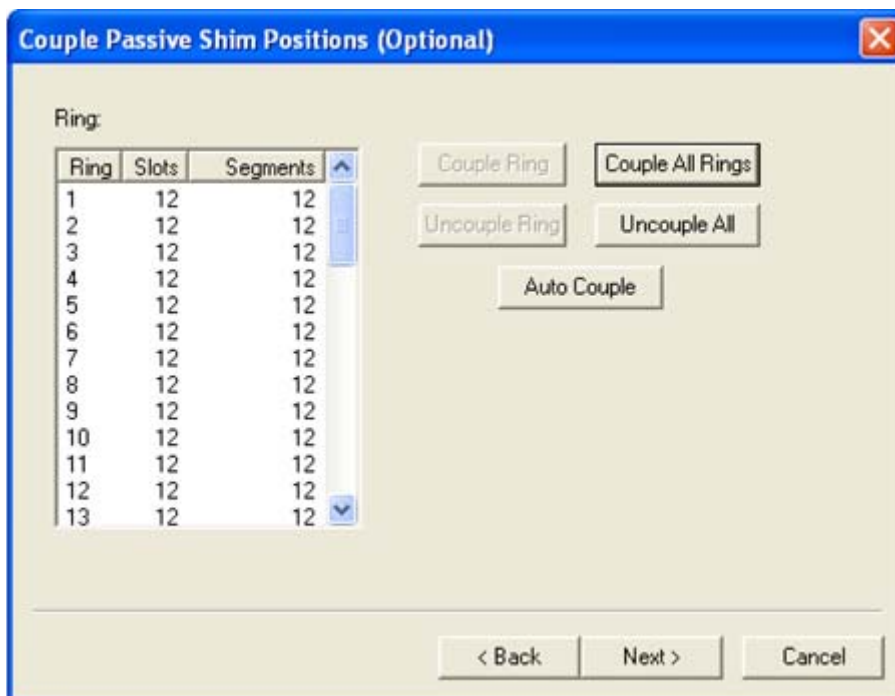
1 - 0 deg.  
2 - 30 deg.  
3 - 60 deg.  
4 - 90 deg.  
5 - 120 deg.  
6 - 150 deg.  
7 - 180 deg.  
8 - 210 deg.  
9 - 240 deg.  
10 - 270 deg.  
11 - 300 deg.  
12 - 330 deg.

Enable All  
Enable Ring  
Disable All  
Disable Ring  
Disable Empty

< Back   Next >   Cancel

14. Click [Next >] without making changes on the *Couple Passive Shim Positions (Optional)* screen when it displays.

Illustration 3-177: Couple Passive Shim Positions (Optional) Screen



15. Check that the **Output File** name on the *Select Shim Target PPM* screen that displays next matches the .map filename, but with a **.shm** or **.cur** extension. Make no other changes on this screen.

**NOTE:** If the output file is not located as stated in [Table 3-23](#) and the full path to the correct map file cannot be recalled, click [Browse], locate the correct .map file, select it, and open the file. Once opened, change the file extension to **.shm** for Passive Shimming output or to **.cur** for S/C Shimming output.

Illustration 3-178: Select Shim Target PPM Screen

Select Shim Target PPM

Virgin Field Statistics

PPM: 362.894195

Average Field: 15010.342368

Post Quantization: 200

Output File: R06783000000D3D.shm

Browse...

< Back Finish Cancel

16. Click [< Back] as needed to review, and verify that all the entries and choices on the previous screens are correct.
17. Click [Finish] to begin processing the shimming information with ShimTool.

**NOTE:** A *Shim Status* window displays while the ShimTool is processing, and closes when processing is completed.

#### 11.4.3.3 Post ShimTool Processing

1. Click [Dismiss] when the *Shim Summary* screen (similar to the one below) displays.

Illustration 3-179: Shim Summary Screen

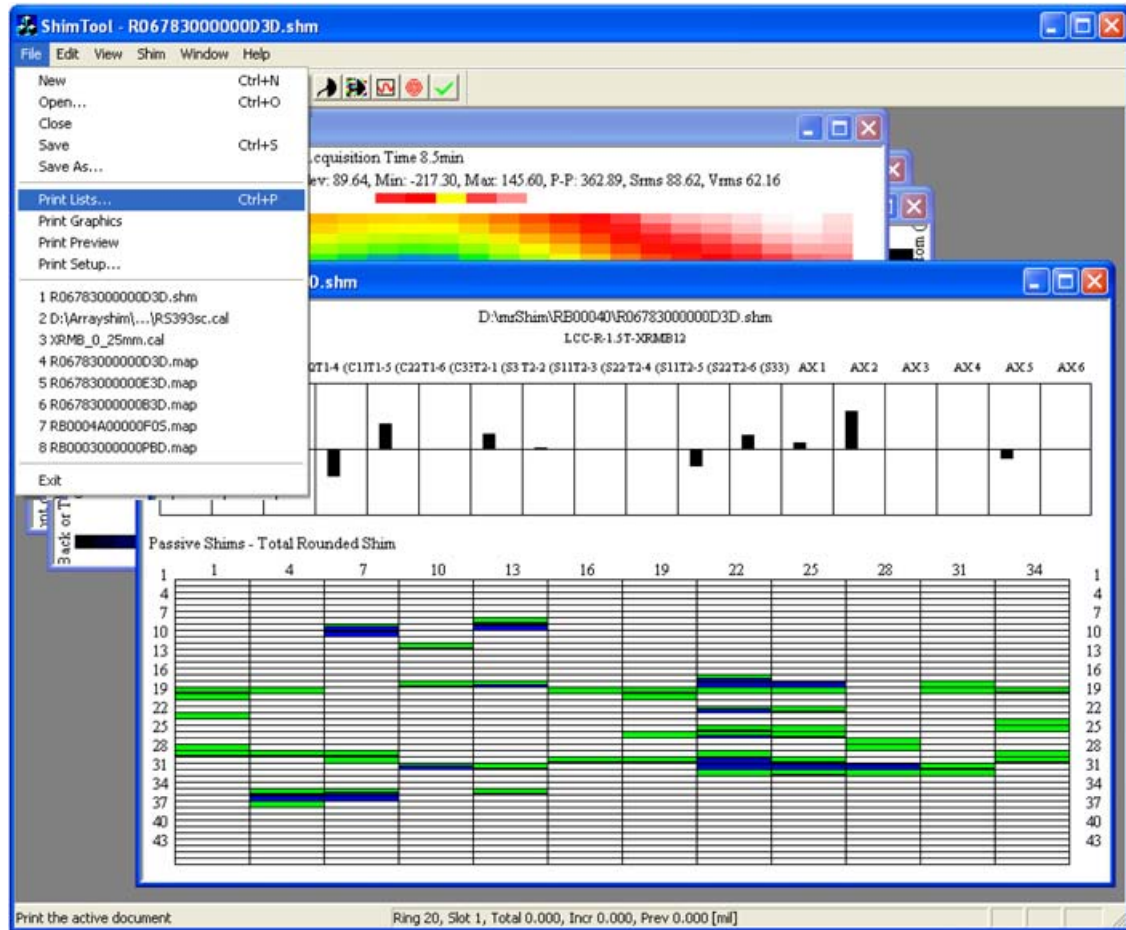
The screenshot shows a software window titled "Shim Summary - R06783000000D3D.shm". It contains several input fields for shim parameters. At the top, "Total Shims" and "Changed Shims" are both set to 61. Below that, "Total Mass [g]" and "Net Mass Change [g]" are both 359.0000. The "Iterations" field is set to 1514. A "Predicted Field" section contains a "Target PPM" of 18.0068. This section is divided into "Unrounded" and "Quantized" sub-sections. The "Unrounded" section shows a "PPM" of 9.4741 and an "Average" of 15025.6326. The "Quantized" section shows a "PPM" of 12.7507 and an "Average" of 15025.6362. A "Dismiss" button is located at the bottom center of the window.

| Parameter           | Value      |
|---------------------|------------|
| Total Shims         | 61         |
| Changed Shims       | 61         |
| Total Mass [g]      | 359.0000   |
| Net Mass Change [g] | 359.0000   |
| Iterations          | 1514       |
| Target PPM          | 18.0068    |
| Unrounded PPM       | 9.4741     |
| Unrounded Average   | 15025.6326 |
| Quantized PPM       | 12.7507    |
| Quantized Average   | 15025.6362 |

2. When the more complete display of the shim iteration results displays (similar to the one below), print the ShimTool results.

**NOTE:** Illustrations show the D: drive (Windows XP) directory paths used by ShimTool installations; Windows 2000 uses the C: drive.

Illustration 3-180: Shim Iteration Results Displays



- The .cur / .shm text file is automatically saved by the ShimTool. Print the file from Notepad, WordPad, or a word processor for pages similar to those shown below. (The ShimTool can also open previously saved files for on-screen viewing by selecting **Show the output SHM file** from the *Shim* menu.)

Illustration 3-181: Sample ShimTool Text File

```

R06783000000D3D.shm - Notepad
File Edit Format View Help
SHIM OUTPUT FILE -- Generated By ShimTool 2.5d

      SHIM MAGNET TYPE : Cylindrical
      MAGNET TYPE      : LCC-R-1.5T-XRMB12
      AUTO-TOLERANCE   : 0.0000 @ 0.0000 PPM
      CRITICAL TARGET  : 0.0000 PPM
      INPUT FIELD FILE  : R06783000000D3D.map
      OUTPUT FILE       : D:\Arrayshim\R067830\R06783000000D3D.shm
      CALIBRATION FILE  : XRMB_0_25mm.cal
ANALYSIS OPTION: ADDITION ONLY (FIRST TIME SHIMMING)
MINIMIZATION TECHNIQUE : ABSOLUTE

                                     ROUNDED
                                     -----
Total Number of Shim Pockets          61
Changed Number of shim Pockets        61
Changed Number of SC Shims            12
Total Passive Shim Mass (mil)         359.0000
Net Passive Shim Mass Change (mil)    359.0000 (+359.0000/0.0000)
Anticipated Field Tolerances (PPM)    12.751

AVERAGE FIELD (Gauss)                 Initial= 15010.34
                                      Final= 15025.64

FIELD TOLERANCE (PPM)
                                      Requested= Unified
                                      Final Before Rounding= 9.474

The mapped field inhomogeneity is ( 653 pts) 362.89
The reported field inhomogeneity is ( 329 pts) 361.75 (Spec <= 14.00 Peak-to-Peak PPM)
OUT OF SPEC

NUMBER OF LP ITERATIONS                 = 1514
RUN COMPLETED Fri Jan 23 14:16:32 2009 (1232738192)

The recomp field inhomogeneity is 361.05 ppm
  48 cm axis: -19.64 to +135.02 (spec <= 10.00 PPM peak-to-peak) OUT OF SPEC
  45 cm dsv : -209.03 to +152.02 (spec +/- 6.00 PPM) OUT OF SPEC
  35 cm dsv : -141.28 to +98.45 (spec +/- 0.80 PPM) OUT OF SPEC
  25 cm dsv : -91.09 to +53.77 (spec +/- 0.40 PPM) OUT OF SPEC

Unified Shim Info: LCC-4vol-393Unified
Target File : D:\zhenyu_Zhang\Projects\Shimming\CALFiles\RS393B.TOL

Unrounded: (Predicted)
  48 cm axis: 6.45PPM (-5.90/+0.55)
  45 cm dsv : 8.44PPM (-4.87/+3.58)
  35 cm dsv : 0.85PPM (-0.41/+0.43)
  25 cm dsv : 0.14PPM (-0.08/+0.06)
Quantized: (Predicted)
  48 cm axis: 6.21PPM (-5.65/+0.57)
  45 cm dsv : 12.75PPM (-7.14/+5.61)
  35 cm dsv : 2.44PPM (-1.17/+1.27)
  25 cm dsv : 1.19PPM (-0.57/+0.62)

```

- To obtain a multi-page paper version of the ShimTool results that look similar to the one below:
  1. Select **Print Lists...** from the *File* menu, and
  2. Click [OK].

Illustration 3-182: Sample ShimTool Printed Report - R Series LCC, Pages 1 and 2

D:\Arrayshim\R067830\R06783000000D3D.shm  
LCC-R-1.5T-XRMB12

Pg. 1 of 4

Shim File : D:\Arrayshim\R067830\R06783000000D3D.shm  
Magnet Type : Cylindrical - LCC-R-1.5T-XRMB12  
Shim Solution Time : Fri Jan 23 14:16:32 2009

Total Number of Shims Positions: 61  
Changed Number of Shims Positions: 61  
Total Shim Mass : 359.0000 mil  
Net Mass Change : 359.0000 (+359.0000/0.0000) mil  
Anticipated Field Inhomogeneity : 0.000 PPM

Input Field F

D:\Arrayshim\R067830\R06783000000D3D.shm  
LCC-R-1.5T-XRMB12

Pg. 2 of 4

Active Shim Currents

| Coil / Supply | Previous | Change   | New      |
|---------------|----------|----------|----------|
| T1-1 (C31)    | 0.000 A  | 0.000 A  | 0.000 A  |
| T1-2 (C11+)   | 0.000 A  | -7.435 A | -7.435 A |
| T1-3 (C22+)   | 0.000 A  | 5.626 A  | 5.626 A  |
| T1-4 (C11-)   | 0.000 A  | -6.513 A | -6.513 A |
| T1-5 (C22-)   | 0.000 A  | 5.810 A  | 5.810 A  |
| T1-6 (C33)    | 0.000 A  | 0.000 A  | 0.000 A  |
| T2-1 (S31)    | 0.000 A  | 3.586 A  | 3.586 A  |
| T2-2 (S11+)   | 0.000 A  | 0.483 A  | 0.483 A  |
| T2-3 (S22+)   | 0.000 A  | 0.000 A  | 0.000 A  |
| T2-4 (S11-)   | 0.000 A  | 0.000 A  | 0.000 A  |
| T2-5 (S22-)   | 0.000 A  | -4.218 A | -4.218 A |
| T2-6 (S33)    | 0.000 A  | 3.187 A  | 3.187 A  |
| AX 1          | 0.000 A  | 1.641 A  | 1.641 A  |
| AX 2          | 0.000 A  | 8.633 A  | 8.633 A  |
| AX 3          | 0.000 A  | -0.111 A | -0.111 A |
| AX 4          | 0.000 A  | 0.000 A  | 0.000 A  |
| AX 5          | 0.000 A  | -2.465 A | -2.465 A |
| AX 6          | 0.000 A  | 0.000 A  | 0.000 A  |

**NOTE:** The multi-page paper version is easier to use while placing shims in the shim trays.

- Passive Shims—Incremental Rounded Shim information on Page 3 and Passive Shims—Total Rounded Shim information on Page 4 of the ShimTool printout ([Illustration 3-182](#) and [Illustration 3-183](#)) is the same information contained under Incremental—Quantized (for Incremental Rounded Shim) and Total—Quantized (for Total Rounded Shim) in the .shm file in [Illustration 3-184](#), which now shows the amount of change and the resulting total at each shim position to be changed).
- The .cur files do **not** contain incremental information for passive shims.

Illustration 3-183: Sample ShimTool Printed Report - RB Series LCC, Pages 1 and 2

Y:\FinalT\RB0004A\RB0004A00000B0N.shm  
LCC-RB-1.5T-XRMB12

Pg. 1 of 4

Shim File : Y:\FinalT\RB0004A\RB0004A00000B0N.shm  
Magnet Type : Cylindrical - LCC-RB-1.5T-XRMB12  
Shim Solution Time : Wed Mar 11 03:40:04 2009

Total Number of Shims Positions: 78  
Changed Number of Shims Positions: 26  
Total Shim Mass : 409.0000 mil  
Net Mass Change : 42.0000 (+43.0000/-1.0000) mil  
Anticipated Field Inhomogeneity : 0.000 PPM

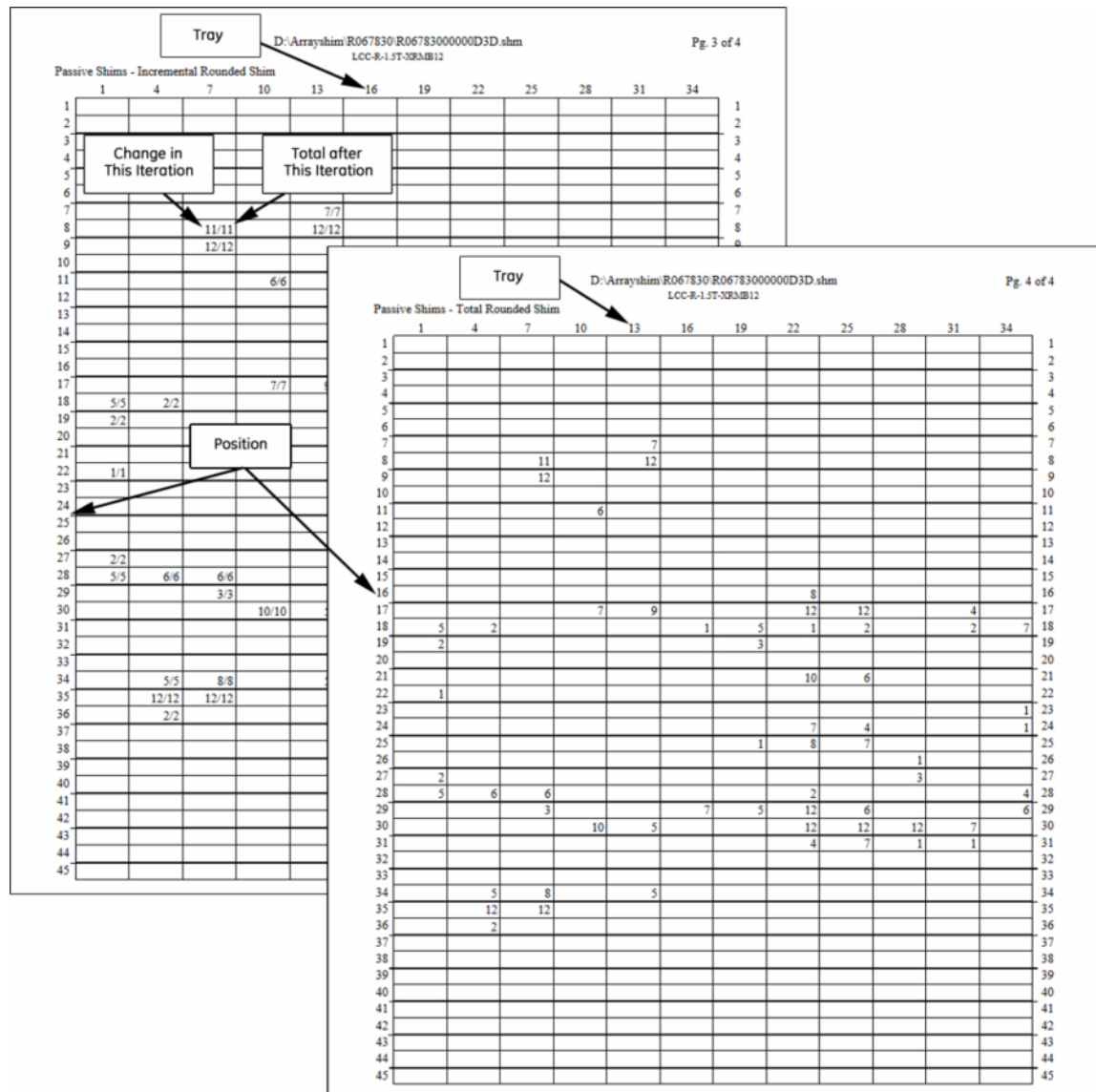
Y:\FinalT\RB0004A\RB0004A00000B0N.shm  
LCC-RB-1.5T-XRMB12

Pg. 2 of 4

Active Shim Currents

| Coil / Supply | Previous  | Change   | New       |
|---------------|-----------|----------|-----------|
| AX1           | -12.785 A | -0.344 A | -13.129 A |
| AX2           | -8.134 A  | -0.067 A | -8.201 A  |
| AX3           | -5.259 A  | -0.525 A | -5.784 A  |
| AX4           | 0.000 A   | 0.000 A  | 0.000 A   |
| AX5           | 0.000 A   | 0.000 A  | 0.000 A   |
| AX6           | 0.000 A   | 0.000 A  | 0.000 A   |
| T1-1 (C31)    | 0.000 A   | 0.000 A  | 0.000 A   |
| T1-2 (C11+)   | -8.159 A  | -0.164 A | -8.323 A  |
| T1-3 (C22+)   | -5.035 A  | -0.006 A | -5.041 A  |
| T1-4 (C11-)   | -7.223 A  | -0.012 A | -7.235 A  |
| T1-5 (C22-)   | -2.214 A  | -0.167 A | -2.381 A  |
| T1-6 (C33)    | 0.000 A   | 0.000 A  | 0.000 A   |
| T2-1 (S31)    | -5.372 A  | 0.000 A  | -5.372 A  |
| T2-2 (S11+)   | 0.000 A   | 0.015 A  | 0.015 A   |
| T2-3 (S22+)   | -7.858 A  | -0.128 A | -7.986 A  |
| T2-4 (S11-)   | 0.000 A   | 0.000 A  | 0.000 A   |
| T2-5 (S22-)   | 0.000 A   | 0.000 A  | 0.000 A   |
| T2-6 (S33)    | 0.000 A   | 0.000 A  | 0.000 A   |

Illustration 3-184: Sample ShimTool Printed Report - Passive Shimming, Pages 3 and 4



**NOTE:** Use these ShimTool results for [Section 11.4.4](#), Passive Shim Installation.

- Record the Anticipated Field Tolerance (PPM) value found in the .cur / .shm file in the Shimming Process Data under **Predicted PPM** for the current iteration.
- Record the Total Passive Shim Mass (shims) value found in the .cur / .shm file in the *Shimming Process Data* under **Passive Shim Mass (mm)** for the current iteration.

**NOTE:** Shim mass (shims) refers to the total number of shims in all shim tray positions.

- On the first iteration (using the centered map and set to calculate S/C Shimming ONLY), compare the four volumes listed under **The recomp field inhomogeneity is ...** ([Illustration 3-181](#)) to the specifications shown in the table below.

**NOTE:** This step corresponds to the diamond-shaped decision box labelled *Are PPM Values For All 4 Volumes Under “Quantized (Predicted)” Within Spec?* in [Illustration 3-148](#), Shimming Flowchart.

**Table 3-24: Volumetric Specifications**

| Volume    | Peak-to-Peak ppm Specification            |
|-----------|-------------------------------------------|
| 48 axial  | <10 ppm (peak to peak)                    |
| 45 cm DSV | within $\pm 6$ ppm (maximum to minimum)   |
| 35 cm DSV | within $\pm 0.8$ ppm (maximum to minimum) |
| 25 cm DSV | within $\pm 0.4$ ppm (maximum to minimum) |

- If **all** four volumes listed under **The recomp field inhomogeneity is ...** are **within** specification, the magnet is shimmed. Continue with [Finalization, Step 1](#)
  - If **any** volume listed under **The recomp field inhomogeneity is ...** is **not** within specification, compare the Quantized (Predicted) values for the four volumes to the specifications shown in [Illustration 3-181](#).
  - If **all** four volumes listed under **Quantized (Predicted)** are **within** specification:
    1. Apply the S/C shim currents called for by the ShimTool in conformance with [Section 11.4.5](#), Superconducting Shimming.
    2. Return to [Section 11.4.3.1](#), Field Map Acquisition, to take a new map and run ShimTool for S/C Shimming ONLY.
    3. Continue with [Step 7](#).
  - If **any** volume listed under Quantized (Predicted) is **not** within specification:
    1. Return to [Section 11.4.3.2](#), Acquiring a Shim File, and run ShimTool for **both** Passive and S/C Shimming.
    2. Continue to the next step.
6. After running ShimTool, set for both Passive and S/C Shimming, compare the *Net Mass Change* and the *Changed Number of Shim Positions* values in the ShimTool reports (summary: [Illustration 3-179](#), printed .cur / .shm files, or ShimTool printout ([Illustration 3-182](#), [Illustration 3-183](#) and [Illustration 3-184](#))). Compare these values to the specifications shown in the table below.

**NOTE:** This step corresponds to the diamond-shaped decision box labelled *Passive Shim Change < 5 Positions?* in [Illustration 3-148](#), Shimming Flowchart.

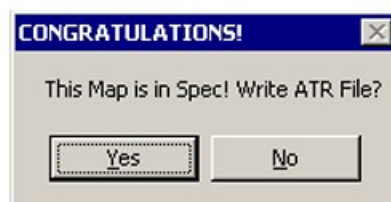
**Table 3-25: Passive Shim Change Specifications**

| Measure                          | Specification |
|----------------------------------|---------------|
| Net Mass Change                  | <5            |
| Changed Number of Shim Positions | <5 Positions  |

- If the specifications in the table above are **not** met (either Net Mass Change is  $\geq 5$  or Changed Number of Shim Positions is  $\geq 5$ ):
    1. Install the passive shims called for by the ShimTool in conformance with [Section 11.4.4](#), Passive Shim Installation.
    2. Once the passive shims are installed, apply the S/C shim currents called for by the ShimTool in conformance with [Section 11.4.5](#), Superconducting Shimming.
    3. Return to [Section 11.4.3.1](#), Field Map Acquisition, to take a new map and run the ShimTool for **both** Passive and S/C Shimming.
    4. Repeat [Step 6](#).
  - If both specifications (Net Mass Change is  $< 5$  and Changed Number of Shim Positions is  $< 5$ ) **are** met:
    1. Return to [Section 11.4.3.2](#), Acquiring a Shim File, to run ShimTool for S/C Shimming ONLY.
    2. Apply the S/C shim currents called for by the ShimTool in conformance with [Section 11.4.5](#), Superconducting Shimming.
    3. Return to [Section 11.4.3.1](#), Field Map Acquisition, to take a new map and run ShimTool for S/C Shimming ONLY.
7. Compare the four volumes in the ShimTool text files listed under **The recomp field inhomogeneity...** ([Illustration 3-181](#)) to the specifications shown in [Table 3-24](#).
- If **any** volume is **not** within specification, return to [Section 11.4.3.1](#), Field Map Acquisition, to take a new map and run ShimTool for S/C Shimming ONLY.
  - If **all** volumes **are** within specification, the screen shown in [Illustration 3-185](#) displays.
    1. Click [No].
    2. Exit the ShimTool and continue with [Finalization, Step 1](#).

**NOTE:** The **Congratulations!** screen indicates the magnet is shimmed to within specification. Writing an ATR file is not needed after the magnet has left the factory.

Illustration 3-185: Congratulations Screen



#### 11.4.4 Passive Shim Installation

##### 11.4.4.1 Shim Tray Removal

1. Do **not** attempt to passive shim a 1.5T magnet without reading, understanding, and following the safety alerts.



### **⚠ DANGER**

POTENTIAL FATAL INJURY!

BEFORE AND WHILE PERFORMING MAGNET PASSIVE SHIMMING:

- MAKE SURE TO REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS, AND
- MAKE SURE TO FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THE SHIMMING PROCEDURE IN THE MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS SUBSECTION OF [CHAPTER 1, MR MAGNET - SAFETY REQUIREMENTS](#).



### **⚠ DANGER**

POTENTIAL FATAL INJURY!

BEFORE STARTING TO PASSIVE SHIM THE MAGNET:

HAVE ALL WORK ASSISTANTS OR WORK OBSERVERS COMPLY WITH THE BUDDY SYSTEM REQUIREMENTS & CERTIFICATION SUBSECTION OF "MR MAGNET - SAFETY REQUIREMENTS."

**WARNING****POTENTIAL PROJECTILE HAZARD!**

FERROMAGNETIC OBJECTS IN A STRONG MAGNETIC FIELD CAN BECOME DANGEROUS PROJECTILES. STEEL SHIMS IN A MAGNETIC FIELD ARE SUBJECT TO A FORCE PROPORTIONAL TO THE STEEL'S MASS AND STRENGTH. IN A 1.5 TESLA FIELD FORCES >50 POUNDS (22.6 KG) MAY BE ENCOUNTERED. AN ATTRACTING FORCE PULLING A SHIM TRAY INTO THE MAGNET WILL BE ENCOUNTERED WHEN A LOADED TRAY IS BROUGHT WITHIN 3 FT. (1 M) OF THE MAGNET BORE. AN ATTRACTING AND/OR REPELLING FORCE WILL BE ENCOUNTERED WHEN INSERTING/REMOVING A LOADED SHIM TRAY INTO/FROM A TRACK INSIDE THE MAGNET BORE. A REPELLING FORCE ON A SHIM TRAY INSIDE THE MAGNET'S BORE MAY PROPEL THE TRAY OUT FROM THE MAGNET WHEN THE TRAY IS RELEASED FROM ITS FIXED POSITION.

MAKE SURE TO FOLLOW THESE SAFETY REQUIREMENTS DURING PASSIVE SHIM INSTALLATION/REMOVAL:

- ASSEMBLE ALL EQUIPMENT AND CHECK THE FUNCTIONALITY IN CONFORMANCE WITH THE PASSIVE SHIM INSTALLATION PROCEDURE,
- LOAD THE SHIM TRAYS OUTSIDE THE MAGNET ROOM AND NEVER LOOSEN THE INDIVIDUAL SHIMS INSIDE THE MAGNET ROOM,
- SECURE THE SHIMS IN THE SHIM TRAYS AND INSTALL THE TRAYS INTO THE MAGNET IN STRICT CONFORMANCE WITH THE PASSIVE SHIM INSTALLATION PROCEDURE,
- WEAR SAFETY GOGGLES/GLASSES DURING SHIM TRAY INSTALLATION/REMOVAL TO PREVENT EYE INJURY FROM SUDDEN TRAY MOVEMENT,
- WEAR LEATHER GLOVES DURING SHIM TRAY INSTALLATION/REMOVAL TO PREVENT SKIN INJURY FROM FIBERGLASS SLIVERS,
- HOLD THE SHIM TRAYS FIRMLY IN PLACE WHEN RELEASING THE INSTALLED TRAYS, AND MAKE SURE ALL LOADED SHIM TRAYS ARE RESTRAINED (BY HAND, SHIM TRAY PULLER, OR FIXED INTO ITS TRACK), ESPECIALLY THE TRAYS MARKED ON THE END FLANGE AS MOUNTED IN AN OUTWARD REPELLING FORCE,
- INSERT/REMOVE THE SHIM TRAYS WITH A GRADUAL MOTION,
- TWO PERSONS ARE REQUIRED TO INSTALL/REMOVE THE TRAYS,
- DO NOT LOAD THE SHIM TRAYS WHILE ANYONE IS DIRECTLY IN LINE WITH THE TRAY ON EITHER END OF THE MAGNET, AND
- NEVER POSITION YOURSELF BETWEEN A SHIM TRAY AND THE MAGNET.
- A CLEAR PATH PER [CHAPTER 1, MR MAGNET - SAFETY REQUIREMENTS](#) IS REQUIRED FOR PASSIVE SHIM REMOVAL AND INSERTION.

**CAUTION**

Potential Skin Injury!

Fiberglass from the shim trays can splinter while inserting/removing trays.

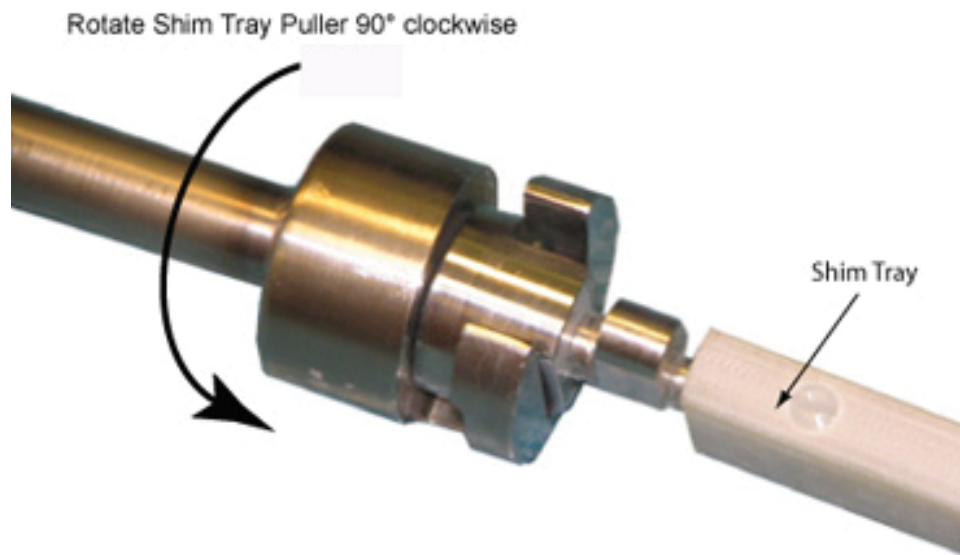
Wear no-slip gloves during shim tray insertion/removal to protect against fiberglass splinters from the trays.

2. Slide the XRMB shim tray puller over the shim tray to be removed, and rotate the puller 90 degrees counterclockwise to secure the connection.

Illustration 3-186: XRMB Shim Tray Puller



Illustration 3-187: Rotating XRMB Shim Tray Puller

**CAUTION**

Potential Projectile Hazard!

Ferromagnetic objects in a strong magnetic field can become dangerous projectiles.

Be prepared to manage the effects of a 1.5T field on the shim tray during tray insertion/removal. Adhere to all safety material listed above.

**CAUTION**

Potential Skin Injury!

Fiberglass from the shim trays can splinter while inserting/removing trays.

Wear no-slip gloves during shim tray insertion/removal to protect against fiberglass splinters from the trays.

3. Carefully pull the shim tray straight out from its track, and have the work assistant hold onto the tray as soon as possible.

Illustration 3-188: Shim Tray Removal



4. Work Assistant: Securely hold the shim tray, and carry the tray and the puller away from the magnet's field to a shim tray loading area in another room.

#### 11.4.4.2 Loading Passive Shims

**NOTICE**

Load shims into the shim tray in a room away from the magnet's field.

1. **First person:** Identify which component shims are required to meet the total shim mass specified by the ShimTool for each position in each tray.

**NOTE:** See [Illustration 3-189](#) for a sample ShimTool shimming report (also shows the D: drive (Windows XP) directory paths used by ShimTool installations; Windows 2000 uses the C: drive), [Illustration 3-190](#) for the LCC shim tray layout, and [Illustration 3-191](#) for assembly of component shim per shim position.

Illustration 3-189: Sample ShimTool Printed Report, Pages 3 and 4

Tray D:\Arrayshim\R067830\R06783000000D3D.shm  
LCC-R-1.5T-3093.B12 Pg. 3 of 4

Passive Shims - Incremental Rounded Shim

|    | 1 | 4 | 7 | 10 | 13 | 16 | 19 | 22 | 25 | 28 | 31 | 34 |
|----|---|---|---|----|----|----|----|----|----|----|----|----|
| 1  |   |   |   |    |    |    |    |    |    |    |    |    |
| 2  |   |   |   |    |    |    |    |    |    |    |    |    |
| 3  |   |   |   |    |    |    |    |    |    |    |    |    |
| 4  |   |   |   |    |    |    |    |    |    |    |    |    |
| 5  |   |   |   |    |    |    |    |    |    |    |    |    |
| 6  |   |   |   |    |    |    |    |    |    |    |    |    |
| 7  |   |   |   |    |    |    |    |    |    |    |    |    |
| 8  |   |   |   |    |    |    |    |    |    |    |    |    |
| 9  |   |   |   |    |    |    |    |    |    |    |    |    |
| 10 |   |   |   |    |    |    |    |    |    |    |    |    |
| 11 |   |   |   |    |    |    |    |    |    |    |    |    |
| 12 |   |   |   |    |    |    |    |    |    |    |    |    |
| 13 |   |   |   |    |    |    |    |    |    |    |    |    |
| 14 |   |   |   |    |    |    |    |    |    |    |    |    |
| 15 |   |   |   |    |    |    |    |    |    |    |    |    |
| 16 |   |   |   |    |    |    |    |    |    |    |    |    |
| 17 |   |   |   |    |    |    |    |    |    |    |    |    |
| 18 |   |   |   |    |    |    |    |    |    |    |    |    |
| 19 |   |   |   |    |    |    |    |    |    |    |    |    |
| 20 |   |   |   |    |    |    |    |    |    |    |    |    |
| 21 |   |   |   |    |    |    |    |    |    |    |    |    |
| 22 |   |   |   |    |    |    |    |    |    |    |    |    |
| 23 |   |   |   |    |    |    |    |    |    |    |    |    |
| 24 |   |   |   |    |    |    |    |    |    |    |    |    |
| 25 |   |   |   |    |    |    |    |    |    |    |    |    |
| 26 |   |   |   |    |    |    |    |    |    |    |    |    |
| 27 |   |   |   |    |    |    |    |    |    |    |    |    |
| 28 |   |   |   |    |    |    |    |    |    |    |    |    |
| 29 |   |   |   |    |    |    |    |    |    |    |    |    |
| 30 |   |   |   |    |    |    |    |    |    |    |    |    |
| 31 |   |   |   |    |    |    |    |    |    |    |    |    |
| 32 |   |   |   |    |    |    |    |    |    |    |    |    |
| 33 |   |   |   |    |    |    |    |    |    |    |    |    |
| 34 |   |   |   |    |    |    |    |    |    |    |    |    |
| 35 |   |   |   |    |    |    |    |    |    |    |    |    |
| 36 |   |   |   |    |    |    |    |    |    |    |    |    |
| 37 |   |   |   |    |    |    |    |    |    |    |    |    |
| 38 |   |   |   |    |    |    |    |    |    |    |    |    |
| 39 |   |   |   |    |    |    |    |    |    |    |    |    |
| 40 |   |   |   |    |    |    |    |    |    |    |    |    |
| 41 |   |   |   |    |    |    |    |    |    |    |    |    |
| 42 |   |   |   |    |    |    |    |    |    |    |    |    |
| 43 |   |   |   |    |    |    |    |    |    |    |    |    |
| 44 |   |   |   |    |    |    |    |    |    |    |    |    |
| 45 |   |   |   |    |    |    |    |    |    |    |    |    |

Change in This Iteration

Total after This Iteration

Position

Tray D:\Arrayshim\R067830\R06783000000D3D.shm  
LCC-R-1.5T-3093.B12 Pg. 4 of 4

Passive Shims - Total Rounded Shim

|    | 1 | 4 | 7 | 10 | 13 | 16 | 19 | 22 | 25 | 28 | 31 | 34 |
|----|---|---|---|----|----|----|----|----|----|----|----|----|
| 1  |   |   |   |    |    |    |    |    |    |    |    |    |
| 2  |   |   |   |    |    |    |    |    |    |    |    |    |
| 3  |   |   |   |    |    |    |    |    |    |    |    |    |
| 4  |   |   |   |    |    |    |    |    |    |    |    |    |
| 5  |   |   |   |    |    |    |    |    |    |    |    |    |
| 6  |   |   |   |    |    |    |    |    |    |    |    |    |
| 7  |   |   |   |    |    |    |    |    |    |    |    |    |
| 8  |   |   |   |    |    |    |    |    |    |    |    |    |
| 9  |   |   |   |    |    |    |    |    |    |    |    |    |
| 10 |   |   |   |    |    |    |    |    |    |    |    |    |
| 11 |   |   |   |    |    |    |    |    |    |    |    |    |
| 12 |   |   |   |    |    |    |    |    |    |    |    |    |
| 13 |   |   |   |    |    |    |    |    |    |    |    |    |
| 14 |   |   |   |    |    |    |    |    |    |    |    |    |
| 15 |   |   |   |    |    |    |    |    |    |    |    |    |
| 16 |   |   |   |    |    |    |    |    |    |    |    |    |
| 17 |   |   |   |    |    |    |    |    |    |    |    |    |
| 18 |   |   |   |    |    |    |    |    |    |    |    |    |
| 19 |   |   |   |    |    |    |    |    |    |    |    |    |
| 20 |   |   |   |    |    |    |    |    |    |    |    |    |
| 21 |   |   |   |    |    |    |    |    |    |    |    |    |
| 22 |   |   |   |    |    |    |    |    |    |    |    |    |
| 23 |   |   |   |    |    |    |    |    |    |    |    |    |
| 24 |   |   |   |    |    |    |    |    |    |    |    |    |
| 25 |   |   |   |    |    |    |    |    |    |    |    |    |
| 26 |   |   |   |    |    |    |    |    |    |    |    |    |
| 27 |   |   |   |    |    |    |    |    |    |    |    |    |
| 28 |   |   |   |    |    |    |    |    |    |    |    |    |
| 29 |   |   |   |    |    |    |    |    |    |    |    |    |
| 30 |   |   |   |    |    |    |    |    |    |    |    |    |
| 31 |   |   |   |    |    |    |    |    |    |    |    |    |
| 32 |   |   |   |    |    |    |    |    |    |    |    |    |
| 33 |   |   |   |    |    |    |    |    |    |    |    |    |
| 34 |   |   |   |    |    |    |    |    |    |    |    |    |
| 35 |   |   |   |    |    |    |    |    |    |    |    |    |
| 36 |   |   |   |    |    |    |    |    |    |    |    |    |
| 37 |   |   |   |    |    |    |    |    |    |    |    |    |
| 38 |   |   |   |    |    |    |    |    |    |    |    |    |
| 39 |   |   |   |    |    |    |    |    |    |    |    |    |
| 40 |   |   |   |    |    |    |    |    |    |    |    |    |
| 41 |   |   |   |    |    |    |    |    |    |    |    |    |
| 42 |   |   |   |    |    |    |    |    |    |    |    |    |
| 43 |   |   |   |    |    |    |    |    |    |    |    |    |
| 44 |   |   |   |    |    |    |    |    |    |    |    |    |
| 45 |   |   |   |    |    |    |    |    |    |    |    |    |

Illustration 3-190: Shim Tray Layout

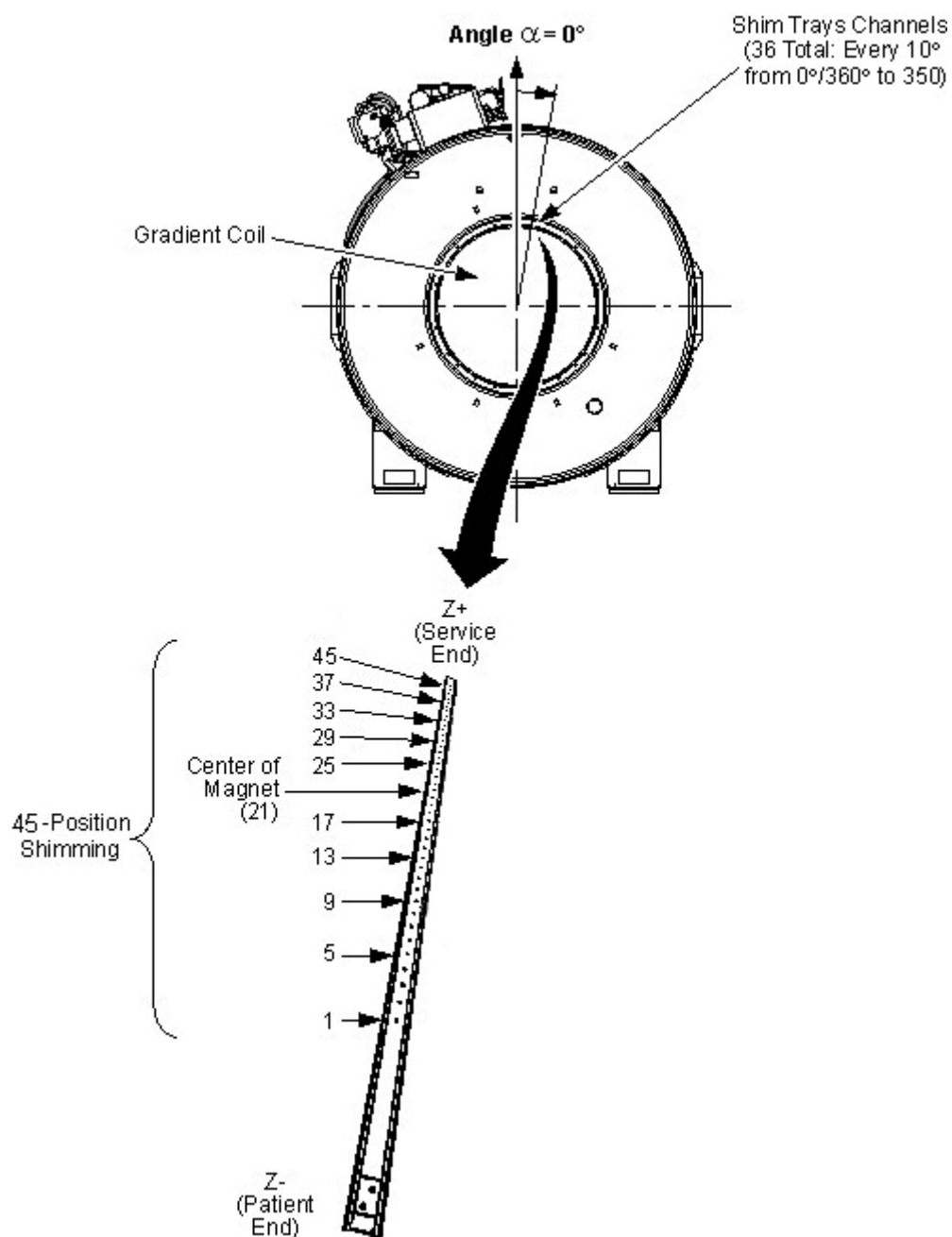
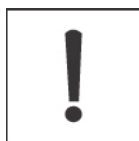
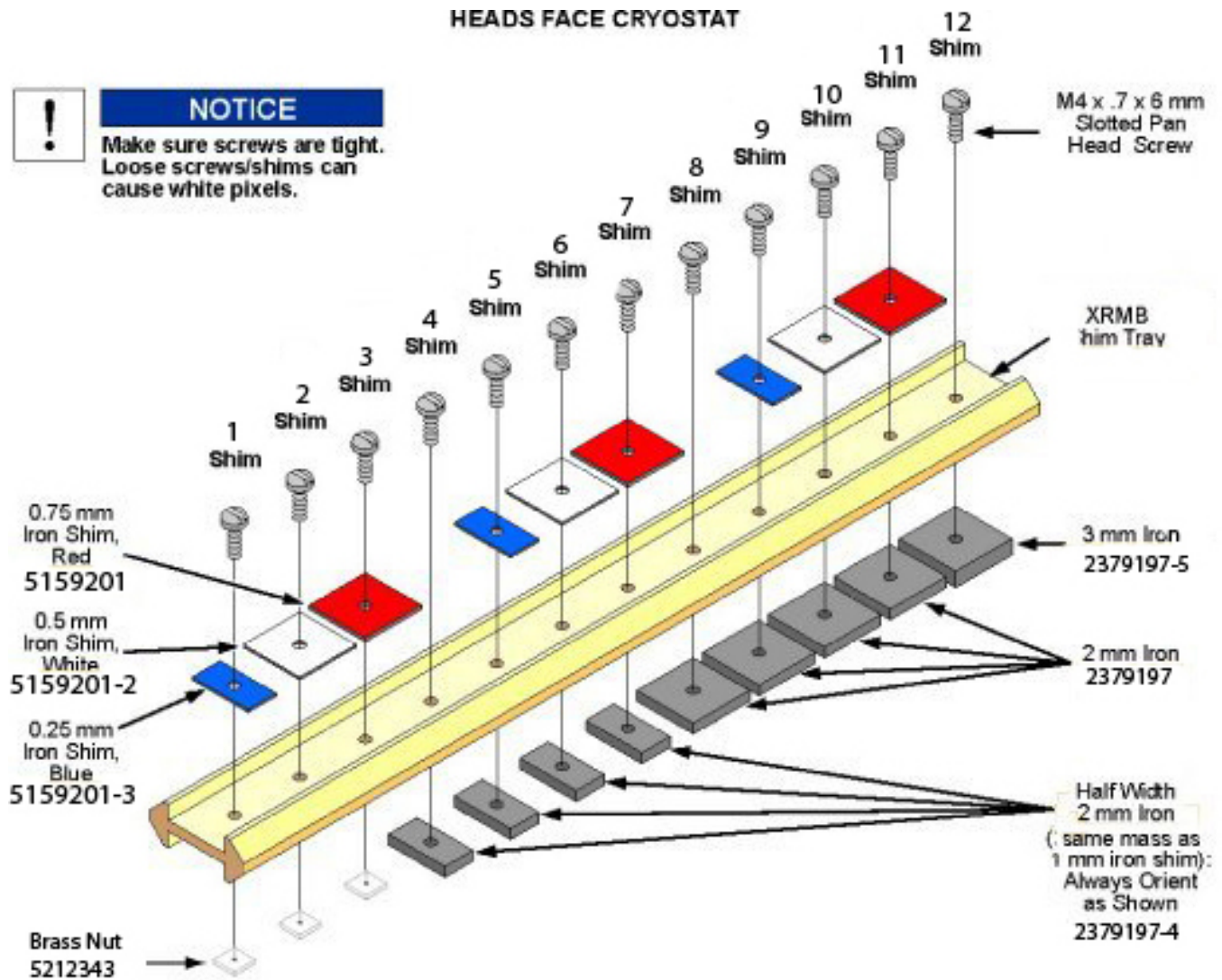


Illustration 3-191: Assembly of Component Shims Per Shim Position

**NOTICE**

Make sure screws holding shims to shim tray are fully tightened using Torque Screwdriver 46-294167P81. Loose screws/shims can cause white pixels.

2. Securely screw the component shim arrangement to the shim tray position as shown in [Illustration 3-191](#).
  - a. Torque each screw to 10 lb-in using Torque Screwdriver 46-294167P81 included in HDv Passive Shimming Kit 5266042.
  - b. Orient every position in each tray identically (screw heads facing the same way, etc.)
3. **Second person (NOT first):** Check that the incremental shims were added/subtracted in conformance with the Passive Shims-Incremental Rounded Shim information on Page 3 of the ShimTool printout ([Illustration 3-189](#)).

**NOTICE**

Do NOT change the name of the .shm file. During the next shimming iteration, the ShimTool uses the steel quantity/location information from this iteration's .shm file and a new map of the field with that steel quantity installed to calculate its new report.

4. **Both people, separately:** Check that the total shims at each tray position equal the total shims for that tray position specified in the Passive Shims–Total Rounded Shim information on Page 4 of the ShimTool printout ([Illustration 3-189](#)).
  - If the actual shim totals in **each** position **match** the .shm file in the ShimTool printout, continue to [Section 11.4.4.3](#), Shim Tray Insertion.
  - If the actual shim totals in **any** position do **not** match the .shm file in the ShimTool printout:
    1. Edit the .shm file in this iteration of Total–Quantized pages to reflect the actual steel on the trays.
    2. Save the edited .shm file using its **existing** name.
    3. Continue to the next section, Shim Tray Insertion.

**11.4.4.3 Shim Tray Insertion****CAUTION**

Potential Skin Injury!

Fiberglass from the shim trays can splinter while inserting/removing trays.

Wear no-slip gloves during shim tray insertion/removal to protect against fiberglass splinters from the trays.

**CAUTION**

Potential Projectile Hazard!

Ferromagnetic objects in a strong magnetic field can become dangerous projectiles.

Be prepared to manage the effects of a 1.5T field on the shim tray during tray insertion/removal. Adhere to all safety material listed above.

**NOTICE**

Reconnect the shim tray puller to the shim tray in a room away from the magnet's field.

1. Securely carry the puller and tray to the magnet.

**CAUTION**

Potential Pinch Hazard!

Fingers can get pinched during tray insertion when the magnetic field pulls the tray into the magnet.

Guide trays into the tracks using a flat hand or fist, not fingers.

2. Begin to carefully slide the shim tray straight into its track using both hands as shown below. Have the Work Assistant help guide the tray into the track.

Illustration 3-192: Hand Guidance of Shim Tray Insertion



3. Slide the XRMB shim tray puller over the shim tray to be inserted.
4. Rotate the puller 90 degrees clockwise to secure the connection as shown above.
5. Fully insert the shim tray into the magnet. (The illustration below shows Shim Tray 6 not fully inserted.)

Illustration 3-193: Shim Tray Not Fully Inserted



6. Rotate the puller 90 degrees counterclockwise to disengage it from the tray.

#### 11.4.5 Superconducting Shimming

1. Do **not** attempt to superconductive shim a 1.5T magnet without reading, understanding, and following the safety alerts and notices.



### **▲ DANGER**

POTENTIAL FATAL INJURY!

BEFORE AND WHILE PERFORMING SUPERCONDUCTING SHIMMING OF THE MAGNET:

- MAKE SURE TO REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS, AND
- MAKE SURE TO FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THE SHIMMING PROCEDURE IN THE MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS SUBSECTION OF [CHAPTER 1, MR MAGNET - SAFETY REQUIREMENTS](#).

**⚠ DANGER**

POTENTIAL FATAL INJURY!

BEFORE STARTING TO S/C SHIM THE MAGNET:

HAVE ALL WORK ASSISTANTS OR WORK OBSERVERS COMPLY WITH THE BUDDY SYSTEM REQUIREMENTS AND CERTIFICATION SUBSECTION OF "MR MAGNET - SAFETY REQUIREMENTS."

**⚠ WARNING**

ASPHYXIATION HAZARD!

THE SHIMMING PROCEDURE GENERATES ODORLESS, COLORLESS HELIUM GAS THAT DISPLACES OXYGEN IN THE AIR.

MAKE SURE MAGNET ROOM VENT EXHAUST FAN IS ON BEFORE STARTING TO S/C SHIM THE MAGNET.

**⚠ CAUTION**

Potential Magnet Quench!

A closed Nupro Valve during Shim Lead engagement can cause a magnet quench. Make sure the Nupro Valve is fully open and the valve is frosted before slowly engaging the Shim Lead. Do NOT remove the Shim Lead's Orifice Cap that controls helium gas flow. Always engage/disengage the Shim Lead in conformance with Shim Lead Engagement and Disengagement.

**NOTICE**

Ensure that the shim power supply is calibrated in conformance with the vendor's manual and that the shim heater currents are set as shown below before use.

- Transverse 1 Shim Heater Current: 610 mA,  $\pm 5$  mA
- Transverse 2 Shim Heater Current: 610 mA,  $\pm 5$  mA
- Axial Shim Heater Current: 710 mA,  $\pm 10$  mA

**NOTICE**

The Recommended Total Current and polarity values calculated by the shim program must be entered into the S/C Correction Coils before performing subsequent field mapping.

The coil groups are:

- Transverse Coils, Odd: T1-1 to T1-6
- Transverse Coils, Even: T2-1 to T2-6
- Axial Coils: AX1 to AX6

Each coil group is connected to one heater circuit on the shim power supply. If the only current change was in the Axial 2 Coil, turning the Axial Switch Heater on would dump the currents in Axial Coils (AX1 to AX6), and the appropriate currents for each Axial Coil would need to be entered.



## NOTICE

Before turning on the shim coil power supply and applying shim currents, make sure the following conditions exist to prevent irreparable damage to the vapor-cooled Shim Leads:

- Orifice on the Shim Lead Nupro Vent Valve is in place.
- Shim Lead is frosting and venting.
- Shim Power Supply current controls are set to zero (fully CCW).



## NOTICE

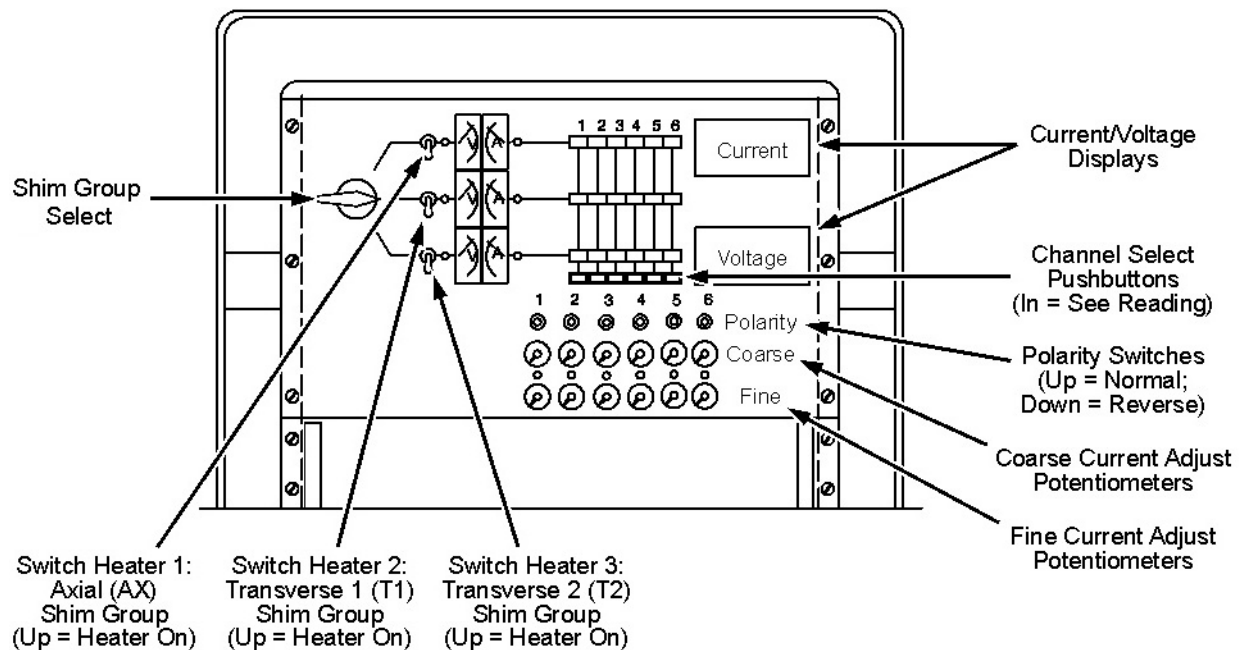
Do not move articles or equipment in the Magnet Room while adjusting shim currents.

2. Make sure the Shim Lead is engaged (Shim Lead Engagement and Disengagement), Shim Lead Nupro Vent Valve is open, vapor is escaping from the Shim Lead and the Lead is frosted, then turn on the power supply.
3. Repeat the sub-steps that follow to input the shim currents for each shim group in the sequence: Transverse 1 (T1), Transverse 2 (T2), and finally Axial as shown in the table below.
  - a. Set the Shim Group Select Switch to the appropriate group: T1, T2, or Axial. (See [Illustration 3-194](#).)

**Table 3-26: S/C Shim Groups**

| Course/Fine Adjust Potentiometer & Polarity Light/Switch | Transverse 1 Shim Group          | Transverse 2 Shim Group | Axial Shim Group      |
|----------------------------------------------------------|----------------------------------|-------------------------|-----------------------|
|                                                          | Shim Group Select Switch Setting |                         |                       |
|                                                          | Middle                           | Bottom                  | Top                   |
|                                                          | Coil (Circuit)                   | Coil (Circuit)          | Coil (Circuit)        |
| 1                                                        | T1-1                             | T2-1                    | AX1 (Z <sup>1</sup> ) |
| 2                                                        | T1-2                             | T2-2                    | AX2 (Z <sup>2</sup> ) |
| 3                                                        | T1-3                             | T2-3                    | AX3 (Z <sup>3</sup> ) |
| 4                                                        | T1-4                             | T2-4                    | AX4 (Z <sup>4</sup> ) |
| 5                                                        | T1-5                             | T2-5                    | AX5 (Z <sup>5</sup> ) |
| 6                                                        | T1-6                             | T2-6                    | AX6 (Z <sup>6</sup> ) |

Illustration 3-194: Shim Power Supply Controls (Front Panel)



**NOTE:** The present currents and polarities, referred to in the following steps, are those that currently exist in each shim coil. They are ShimTool calculated/reported in the *Previous* column under Active Shim Currents in [Illustration 3-195](#) or [Illustration 3-196](#) in its previous iteration, and loaded into the shim coils and recorded in the *New* column in the SC Shimming Data.

[Illustration 3-195](#) and [Illustration 3-196](#) show the D: drive (Windows XP) directory paths used by ShimTool installations; Windows 2000 uses the C: drive.

Illustration 3-195: Sample ShimTool Printed Report - R Series LCC, Pages 1 and 2

D:\Arrayshim\R067830\R06783000000D3D.shm  
LCC-R-1.5T-XRMB12

Pg. 1 of 4

Shim File : D:\Arrayshim\R067830\R06783000000D3D.shm  
Magnet Type : Cylindrical - LCC-R-1.5T-XRMB12  
Shim Solution Time : Fri Jan 23 14:16:32 2009

Total Number of Shims Positions: 61  
Changed Number of Shims Positions: 61  
Total Shim Mass : 359.0000 mil  
Net Mass Change : 359.0000 (+359.0000/0.0000) mil  
Anticipated Field Inhomogeneity : 0.000 PPM

Input Field F

D:\Arrayshim\R067830\R06783000000D3D.shm  
LCC-R-1.5T-XRMB12

Pg. 2 of 4

Active Shim Currents

| Coil / Supply | Previous | Change   | New      |
|---------------|----------|----------|----------|
| T1-1 (C31)    | 0.000 A  | 0.000 A  | 0.000 A  |
| T1-2 (C11+)   | 0.000 A  | -7.435 A | -7.435 A |
| T1-3 (C22+)   | 0.000 A  | 5.626 A  | 5.626 A  |
| T1-4 (C11-)   | 0.000 A  | -6.513 A | -6.513 A |
| T1-5 (C22-)   | 0.000 A  | 5.810 A  | 5.810 A  |
| T1-6 (C33)    | 0.000 A  | 0.000 A  | 0.000 A  |
| T2-1 (S31)    | 0.000 A  | 3.586 A  | 3.586 A  |
| T2-2 (S11+)   | 0.000 A  | 0.483 A  | 0.483 A  |
| T2-3 (S22+)   | 0.000 A  | 0.000 A  | 0.000 A  |
| T2-4 (S11-)   | 0.000 A  | 0.000 A  | 0.000 A  |
| T2-5 (S22-)   | 0.000 A  | -4.218 A | -4.218 A |
| T2-6 (S33)    | 0.000 A  | 3.187 A  | 3.187 A  |
| AX 1          | 0.000 A  | 1.641 A  | 1.641 A  |
| AX 2          | 0.000 A  | 8.633 A  | 8.633 A  |
| AX 3          | 0.000 A  | -0.111 A | -0.111 A |
| AX 4          | 0.000 A  | 0.000 A  | 0.000 A  |
| AX 5          | 0.000 A  | -2.465 A | -2.465 A |
| AX 6          | 0.000 A  | 0.000 A  | 0.000 A  |

Illustration 3-196: Sample ShimTool Printed Report, RB Series LCC, Pages 1 and 2

Y:\FinalT\RB0004A\RB0004A00000B0N.shm  
LCC-RB-1.5T-XRMB12

Pg. 1 of 4

Shim File : Y:\FinalT\RB0004A\RB0004A00000B0N.shm  
Magnet Type : Cylindrical - LCC-RB-1.5T-XRMB12  
Shim Solution Time : Wed Mar 11 03:40:04 2009

Total Number of Shims Positions: 78  
Changed Number of Shims Positions: 26  
Total Shim Mass : 409.0000 mil  
Net Mass Change : 42.0000 (+43.0000/-1.0000) mil  
Anticipated Field Inhomogeneity : 0.000 PPM

Y:\FinalT\RB0004A\RB0004A00000B0N.shm  
LCC-RB-1.5T-XRMB12

Pg. 2 of 4

Active Shim Currents

| Coil / Supply | Previous  | Change   | New       |
|---------------|-----------|----------|-----------|
| AX1           | -12.785 A | -0.344 A | -13.129 A |
| AX2           | -8.134 A  | -0.067 A | -8.201 A  |
| AX3           | -5.259 A  | -0.525 A | -5.784 A  |
| AX4           | 0.000 A   | 0.000 A  | 0.000 A   |
| AX5           | 0.000 A   | 0.000 A  | 0.000 A   |
| AX6           | 0.000 A   | 0.000 A  | 0.000 A   |
| T1-1 (C31)    | 0.000 A   | 0.000 A  | 0.000 A   |
| T1-2 (C11+)   | -8.159 A  | -0.164 A | -8.323 A  |
| T1-3 (C22+)   | -5.035 A  | -0.006 A | -5.041 A  |
| T1-4 (C11-)   | -7.223 A  | -0.012 A | -7.235 A  |
| T1-5 (C22-)   | -2.214 A  | -0.167 A | -2.381 A  |
| T1-6 (C33)    | 0.000 A   | 0.000 A  | 0.000 A   |
| T2-1 (S31)    | -5.372 A  | 0.000 A  | -5.372 A  |
| T2-2 (S11+)   | 0.000 A   | 0.015 A  | 0.015 A   |
| T2-3 (S22+)   | -7.858 A  | -0.128 A | -7.986 A  |
| T2-4 (S11-)   | 0.000 A   | 0.000 A  | 0.000 A   |
| T2-5 (S22-)   | 0.000 A   | 0.000 A  | 0.000 A   |
| T2-6 (S33)    | 0.000 A   | 0.000 A  | 0.000 A   |

- b. Match the Shim Power Supply current and polarity of each channel (coil) in each group to the present current and polarity in the selected shim coil group. (See [Illustration 3-194](#).)

- Set the Polarity Toggle Switches: Up = normal/positive polarity; Down = reversed/negative polarity (as required to match the present polarity for each coil in the group).
- Set the Course/Fine Current Adjust potentiometers on the power supply to match the present current level for each coil in the group.

**NOTE:** The polarity relays do **not** change polarity if there is any current flowing through their respective channels. Reduce the current to zero (0) before the relays can change to the position indicated by the external switches on the power supply.



## NOTICE

Magnet damage is possible if heater currents are not set correctly.



## NOTICE

When the Switch Heaters are turned on, any existing currents in the Shim Coils are discharged into the power supply. To prevent dumping excessive currents through the Shim Leads, match the existing shim currents at the power supply with those in the magnet before turning on the Switch Heaters. Adjust the current to the required new levels after activating the Switch Heaters.

- c. Check that all shim power supply currents/polarities match those in the selected shim coil group.
- d. Turn on the selected group's Switch Heater (T1, T2, or Axial), and wait 30 seconds for the switches to go resistive.

**NOTE:** If there is a polarity change from the present to the new current, set the current to zero (0) before changing the polarity switch.

- e. Set the new Total currents and polarities for the shim coil group using the ShimTool's latest *Active Shim Currents* values (listed in the column labelled *New* on the Active Shim Currents page of the ShimTool report shown in [Illustration 3-195](#)).
- f. Allow the shim currents to settle for the time shown in the table below.

**Table 3-27: Settling Times for New Shim Currents After Re-Establishing ATR**

| Shim Coil Group                                                                                                                                                                   | Settling Time                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| T1                                                                                                                                                                                | 10 minutes (first iteration)                  |
| T2                                                                                                                                                                                | 5 minutes (additional iterations)             |
| Axial*                                                                                                                                                                            | 20 minutes* (first and additional iterations) |
| * Axial times represent minimum times. Axial coupling may require longer wait times to meet the magnetic field stabilization requirements stated in <a href="#">Step 5</a> below. |                                               |

- g. Verify the new currents, then turn off the heater group and wait five minutes.
- h. After waiting five (5) minutes, gradually turn the power supply Current Adjust Potentiometers down to zero (0) Amps.

**NOTE:** If repeated iterations of the shim program call for the same current increment in a shim coil (indicating the Shim Switch is not going resistive), one of the two following actions may help drive the Shim Switch resistive:

- Make sure the Heater current is set properly, **or**
- Set 20 Amps in the coil, then lower the current to the required value.

Adjust slightly as required; 1 mA high is better than 1 mA low.

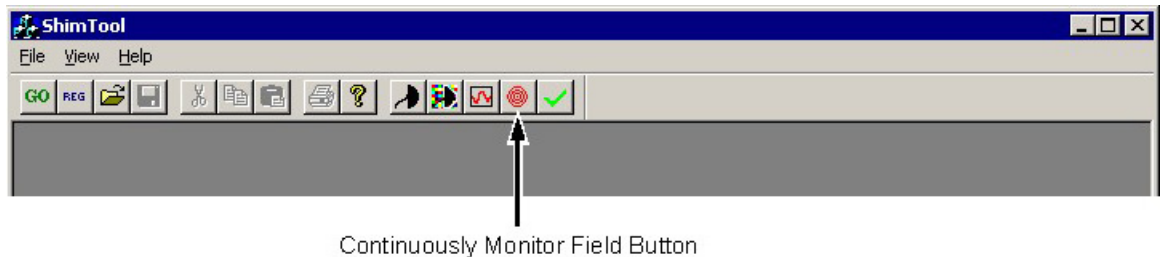
Failure to accurately set the power supply currents and polarity can greatly increase the number of plotting iterations (i.e., time to shim the magnet).

4. Make sure the Field Camera is in the 0° position.
5. **(For Axial Currents Only)**  
**(For Axial Currents Only)** Monitor the field drift until it stabilizes (i.e., <2 Hz change in total field over a two-minute period).

**NOTE:** In some cases, the frequency could vary as much as 100 Hz. Shutting off the Coldhead Compressor during acquisition/mapping of shim data may help reduce the frequency change.

- a. Click [Continuously Monitor Field] (2nd button from right) on the *ShimTool*/main screen.

**Illustration 3-197: Continuously Monitor Field Button on ShimTool Main Screen**



- b. Select **Probe 22**.
  - c. Click [Start] on the screen that displays to begin measuring field strength and drift. Make a note of the starting frequency for use in the next step.
  - d. Wait two minutes, click [Stop], and record the change between the starting and present frequencies.
    - If the change is <2 Hz, proceed to the next step.
    - If the change is ≥2 Hz, repeat [Step 5.c](#) and [Step 5.d](#).
6. Once the magnetic field is stable, turn off the last Switch Heater that was turned on, and allow the Heater to cool for five (5) minutes.

7. Slowly turn all the Course/Fine Current Adjust Potentiometers on the Shim Power Supply back to zero (0) Amps (fully CCW).

**CAUTION**

Potential Quench Hazard!

The axial shim currents may quench if the Shim Switch is in the Axial position when the Shim Power Supply is turned off.

Make sure the Shim Power Supply's Shim Switch is in the Transverse 1 or Transverse 2 position before turning it off.

8. When Superconducting Shimming is complete:
  - a. Turn off the power supply.

**WARNING**

ELECTRICAL SHOCK HAZARD!

CONTACT WITH CONNECTORS LEADING TO AN ENERGIZED POWER SUPPLY CAN CAUSE ELECTRICAL SHOCK.

FOLLOW LOTO PROCEDURES WHILE MAKING ELECTRICAL CONNECTIONS TO THE SHIM POWER SUPPLY.

- b. Disconnect and LOTO input power to the Shim Coil Power Supply. Check that no voltage is present by using a DVM or equivalent measuring device.
  - c. Disconnect all leads between the magnet and the Shim Coil Power Supply (P2, P2-W, Heater Cable).
9. Close the Shim Lead Nupro Vent Valve.

**NOTICE**

Make sure the cap is replaced and does not leak, resulting in gaseous helium loss or frosting.

10. Retract the Shim Lead in conformance with Shim Lead Engagement and Disengagement, tighten the brass cap, and check for leaks.

## 11.5 Finalization

1. Make sure the shimming results are available to other personnel for future shimmming:
  - a. Copy all .map, .cur and .shm files generated during shimmming to a CD,
  - b. Paste the .map, .cur and .shm files to the console,
  - c. Record the directory path of the files pasted to the console in the Shim Process Log of the Shimmming Process Data, and
  - d. Leave the CD on site with this Magnet and Cryogens Subsystem manual.

2. Return the magnet to its normal operating condition in conformance with [Preparations for Field Measurement](#) and the appropriate Service Methods documentation.

**NOTE:** See Field Adjustment after Shimming for field adjustments after shimming and Field Stability Check for magnetic field stability checks.